

SPATIAL ANALYSIS OF CHILD MALNUTRITION IN RWANDA: A District-Level Integration of DHS Data and Agro-Climatic Indicators

By

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DECLARATION

I declare that this thesis entitled *Spatial Analysis of Child Malnutrition in Rwanda* is my original work and has not been submitted to any other institution for the award of a degree or any other academic qualification.

All sources of information used in this study have been properly acknowledged and referenced. The analysis, interpretations, and conclusions presented in this thesis are my own, except where otherwise stated.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AIC	Akaike Information Criterion
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station Data
CI	Confidence Interval
DHS	Demographic and Health Survey
ESDA	Exploratory Spatial Data Analysis
FDR	False Discovery Rate
GADM	Global Administrative Areas Database
GAM	Generalised Additive Model
GIS	Geographic Information System
GLMM	Generalised Linear Mixed Model
GPS	Global Positioning System
ICC	Intraclass Correlation Coefficient
LISA	Local Indicators of Spatial Association
LRT	Likelihood Ratio Test
MARS	Multivariate Adaptive Regression Splines
MODIS	Moderate Resolution Imaging Spectroradiometer
NDVI	Normalised Difference Vegetation Index
OLS	Ordinary Least Squares
OR	Odds Ratio
PCA	Principal Component Analysis
RDHS	Rwanda Demographic and Health Survey
SEM	Spatial Error Model
SLM	Spatial Lag Model
VIF	Variance Inflation Factor
WHO	World Health Organization

Abstract

Child malnutrition remains a major public health challenge in Rwanda and continues to affect child growth, health, and development. According to the Rwanda Demographic and Health Survey (RDHS) 2019/2020, approximately 33% of children under five years were stunted, 2% were wasted, and 8% were underweight, indicating a continuing burden of undernutrition. This study aimed to investigate the spatial distribution and determinants of child malnutrition in Rwanda using the 2019/2020 Rwanda Demographic and Health Survey (RDHS), CHIRPS rainfall data, MODIS Normalized Difference Vegetation Index (NDVI) data, altitude information, and socio-demographic characteristics. The study focused on three indicators of child malnutrition: stunting, wasting, and underweight. District-level prevalence rates were estimated across Rwanda's 30 districts, and exploratory spatial data analysis was conducted using Global Moran's I and Getis-Ord G_i^* hotspot analysis to examine spatial clustering patterns. Environmental variables including NDVI, rainfall anomaly, rainfall variability, and altitude were assigned to each child according to district of residence. Child-level multilevel logistic regression models were fitted using 3,785 children nested within 498 survey clusters to assess the effects of environmental, household, maternal, and child characteristics on malnutrition outcomes. The spatial analysis showed that child malnutrition was not randomly distributed across Rwanda. Stunting hotspots were mainly identified in western districts such as Nyabihu, Rubavu, and Ngororero, while wasting and underweight hotspots were observed mainly in southern districts including Huye and Gisagara. The multilevel analysis showed that rainfall anomaly and altitude remained significant predictors of stunting at the individual level, while no environmental predictor was significant for underweight after controlling for individual and household characteristics. Higher altitude increased the likelihood of stunting, while increased rainfall levels were associated with a lower stunting risk. Household wealth, maternal education, maternal nutritional status, and antenatal care visits were associated with reduced odds of malnutrition, while female children generally showed a lower malnutrition risk compared with male children. Intraclass correlation results further indicated that most variation in stunting occurred at the individual child level, whereas wasting showed stronger clustering at the community level. Overall, the findings suggest that child malnutrition in Rwanda is influenced by a combination of environmental, socio-economic, maternal, and community-level factors. In regard to wasting, the correlation was 0% for the district level intraclass, having 80% of variance concentrated at the survey cluster level. This clearly indicates that wasting reflects acute local conditions rather than district level environmental conditions. Integrating spatial analysis with multilevel modeling provides a more comprehensive understanding of both geographic patterns and individual risk factors and may support targeted nutrition policies and interventions.

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1. Introduction

1.1 Background

Child malnutrition is the one of the most damaging problems facing low-income countries to day. Its consequences reach far beyond the early years of life, shaping a child's capacity to learn, grow, and eventually contribute to society as a productive adult (World Bank, 2018; Victora et al., 2008). Malnutrition is not simply a matter of going hungry; it equally encompasses diets that are poor in essential nutrients, environments that expose young children to repeated infection, and inadequate care practices during the most sensitive window of development (Black et al., 2008). Addressing it demands coordinated action across health, agriculture, education, and social protection.

Globally, the number of stunted children under five dropped from around 203.6 million in 2000 to 149.2 million by 2020 (UNICEF et al., 2020). Stunting, measured as low height-for-age, reflects prolonged nutritional deprivation that can begin before birth (World Health Organization, 2006). Wasting, or low weight-for-height, signals a more recent and acute deterioration in nutritional status, while underweight captures a child's overall weight deficit relative to age and thus embeds elements of both chronic and acute malnutrition (Caulfield et al., 2006). Each indicator tells a distinct story about a child's nutritional history (World Bank, 2018; Sarmiento et al., 2014).

Sub-Saharan Africa carries the heaviest burden and has made the least progress. Unlike other regions where stunting numbers have fallen, the absolute count of stunted children in sub-Saharan Africa actually increased from about 54.4 million in 2000 to 61.4 million by 2020. A comprehensive review of 49 studies in the region by Akombi et al. (2017) identified low maternal education, household poverty, limited access to clean water, low birth weight, and child age and sex as the most consistently documented risk factors for all three malnutrition outcomes. Kandala et al. (2011) similarly found that children from rural area, those born to young or underweight mothers, and those from vulnerable households face disproportionately elevated malnutrition risk. These patterns collectively point to poverty, structural inequality, and service deprivation as the underlying forces driving the regional burden.

Rwanda is one of the few countries in sub-Saharan Africa that has made genuinely impressive progress. It met with the Millennium Development Goal 1, cutting its proportion of underweight children by half between 1990 and 2015 (Mukabutera et al., 2016). National stunting prevalence fell from 44% in 2010 to 33% in 2020, reflecting the tangible impact of government nutrition programmes and policy reforms (Habyarimana et al., 2016). Nevertheless, a 33% stunting rate remains above both the national target of 19% and the African regional average of 30.7%, signalling that much more work lies ahead. To accelerate this progress, successive policy frameworks have set increasingly ambitious reduction targets: 10% by 2030, 5.5% by 2035, and ultimately 3% by 2050 under Rwanda's Vision 2050 (Ministry of Gender and Family, 2024). Looking at those numbers honestly, it is clear that the programmes that produced progress so far are not going to be enough to reach these goals (Binagwaho et al., 2020).

Research within Rwanda has identified several layers of risk. At the child level, being male and

being born underweight below 2.5 kilograms both raise the odds of stunting, wasting, and underweight independently of other factors (Sinha et al., 2018). At the mother's level, limited schooling and low body weight increase the chance that her child will also be malnourished, showing how nutritional disadvantage passes from one generation to the next (Javid and Pu, 2020). Adequate antenatal care has been shown to be protective, with higher numbers of antenatal visits linked to lower stunting rates (Torlesse et al., 2016). At the household level, poverty and reliance on unsafe water sources create additional risk by facilitating diarrhoeal and infectious disease that interferes with nutrient absorption (Getaneh et al., 1998).

Another critical aspect is the geographic nature of malnutrition in Rwanda, although it has not received as much attention. The prevalence of childhood stunting across districts as observed in the 2019/20 Rwanda Demographic and Health Survey varies between 10.7% in Kicukiro to 50.5% in Ngororero (Habimana and Biracyaza, 2019) a disparity of five times thereby illustrating the drawbacks in using national averages for decision-making purposes (Niragire et al., 2025). Spatial analysis has demonstrated that high-prevalence districts exhibit geographical concentration in terms of clustering, especially within the Western and Northern provinces, compared to Eastern province, which has experienced low spatial effects (Uwiringiyimana et al., 2022).

Altitude adds another geographic layer to this picture. Children residing at elevations above 2,000 metres carry a consistently higher stunting risk, possibly because high-altitude physiology, including reduced oxygen availability, can interfere with normal growth trajectories (Baye and Hirvonen, 2020; Niermeyer et al., 2009). Beyond altitude, Rwanda's five distinct agro-ecological zones generate substantial variation in rainfall, soil quality, and crop diversity across districts, all of which bear directly on what food families can access and afford (Haile et al., 2016). Yet the interplay between these agro-climatic conditions and child malnutrition outcomes has not been rigorously examined using spatially explicit methods, leaving a meaningful gap that this study seeks to address (Akombi et al., 2017).

1.2 Problem Statement

Although Rwanda's notable progress in reducing child malnutrition over the past two decades, the national stunting rate stood at 33% in 2020, still well above the national target of 19% and the African average of 30.7%. More critically, this national figure hides big differences between districts: stunting prevalence ranged from 10.7% in Kicukiro district to 50.5% in Ngororero district in the 2019/20 Rwanda Demographic and Health Survey (RDHS) underscoring that aggregate national statistics are insufficient to guide targeted interventions (Niragire et al., 2025).

Existing studies on child malnutrition in Rwanda have relied predominantly on non-spatial analytical methods that fail to capture geographic clustering or district-level heterogeneity (Habyarimana et al., 2016); (Mukabutera et al., 2016). Although (Niragire et al., 2025) advanced the field by applying geo-additive Bayesian models across three RDHS rounds and found significant district-level spatial variation, their study did not integrate agro-climatic indicators such as rainfall patterns, vegetation cover, or drought vulnerability as explanatory variables. Altitude was the only environmental factor included. Similarly, the systematic review by (Akombi et al., 2017) across 49

sub-Saharan African studies did not incorporate spatial methods or examine how agro-climatic factors interact geographically with nutritional outcomes.

Rwanda has different agro-ecological zones, which means that food production is not the same in all districts. This affects crop yields, the types of food people eat, and how much food is available in households. All these factors are important for child nutrition, but many studies have not fully looked at them using spatial methods, from Rwanda's National Nutrition Policy 2019–2024 encourages nutrition interventions that are adapted to each district and involve different sectors. However, there is still a lack of spatial evidence at the district level to support this approach. Because of this, there is a gap in understanding how agro-climatic factors together with social and demographic factors influence the distribution of child malnutrition across the 30 districts of Rwanda. Filling this gap in this study is important for designing effective nutrition programs that are well targeted to specific districts.

1.3 Objectives

1.3.1 General Objective. The general objective of this study is to examine the spatial patterns and multilevel determinants of under-five child malnutrition in Rwanda by analysing district-level geographic variation alongside child-level risk factors.

1.3.2 Specific Objectives.

1. To map the district-level spatial distribution of stunting, wasting, and underweight among children under five in Rwanda using the 2019/20 Rwanda Demographic and Health Survey.
2. To assess spatial clustering of child malnutrition across Rwanda's 30 districts using Global Moran's I , and Getis–Ord G_i^* hotspot analysis.
3. To examine the association between district-level agro-climatic and socio-demographic factors and district-level child malnutrition prevalence using OLS and spatial regression models, and to complement this with child-level multilevel logistic regression analysis.
4. To assess whether district-level environmental factors remain associated with child malnutrition after controlling for child, household, and maternal characteristics using multilevel logistic regression.

1.4 Justification

However, the distribution of child malnutrition in Rwanda is uneven, some districts are heavily burdened while others have less prevalence but few studies have explored the role of environmental factors in the creation of this disparity. Earlier studies have established socio-economic and demographic determinants, but there exists inadequate knowledge regarding the role of environmental factors like precipitation, vegetation status, and elevation in explaining this unevenness in malnutrition. The National Nutrition Policy of Rwanda for 2019–2024 emphasizes locally adapted and multi-sectoral strategies, but lacks sufficient spatial data at the district level to inform policy development. This paper attempts to bridge this gap by providing spatial and child-level analysis of factors associated with child malnutrition and their use in developing nutrition policies.

1.5 Organisation of the Study

This thesis is organized into five chapters. The first chapter defines the purpose of the research along with the problem, the objectives of the research and reasons behind conducting the study. The second chapter gives a review of relevant literature on malnutrition among children, techniques of spatial analysis and environment-based determinants of nutrition. The third chapter discusses about the data that will be used to conduct this research, method of construction of variables, and methodology to conduct the analysis. Then the fourth chapter will include descriptive analysis, spatial analysis, regression analysis and multilevel regression analysis, Then fifth chapter will discuss the policy implications, acknowledges study limitations, and proposes directions for future research

2. Literature Review

2.1 Definitions of Key Concepts

2.1.1 Malnutrition. Malnutrition, which is otherwise called nutritional inadequacy, occurs due to the fact that one's diet is unable to provide the body's physiological needs in terms of normal functioning and growth. It can be categorized into three main types of nutritional inadequacies such as under-nutrition, over-nutrition, and the third condition whereby the person consumes enough amount of calories but the body still lacks certain micronutrients required by the body (Ministry of Gender and Family, 2024). Therefore, malnutrition is not just the problem of lack of sufficient amount of food but also the deficiency of certain elements in the food consumed by the body.

Nutritional deficiency disorders among young children less than five years of age have extremely harmful effects on account of the rapid growth taking place at this stage. Poor nutritional intake during early childhood could impair cognitive development, stunted growth, and weaken immunity against diseases (World Bank, 2018). Such adverse impacts of poor nutrition among children are likely to affect their development well into adulthood; hence it is imperative for governments to prioritise the provision of adequate nutrition to children (Ministry of Gender and Family, 2024). Nutritional deficiency among children is measured in this study using stunting, wasting, and underweight indices.

2.1.2 Stunting. Linear growth failure or stunting takes place when there is an abnormal low stature in a child that corresponds to their chronological age; hence, indicating exposure to nutrition deficiency that is not a result of a short-lived problem in the diet (Akombi et al., 2017). The process starts during fetal development when the nutrition level of the mother is low, and continues up to two years in cases where there is lack of good diet and proper childcare (Ministry of Gender and Family, 2024). Stunting, according to the World Health Organization's definition, refers to a situation when a child's height is more than two standard deviations below the median reference stature (Niragire et al., 2025).

2.1.3 Wasting. A wasting condition is a child who has a lower weight for his/her stature. While stunting is chronic, wasting results from a current deficiency of nutrition. According to the WHO standards, wasting in children occurs when their weight-for-height z-scores lie below negative two standard deviations from the median value of weight-for-height measurements (Niragire et al., 2025). The onset of the condition is attributed to sudden shortages of food, like those caused by drought leading to crop failure. Other causes include illnesses such as diarrhea and respiratory illnesses, which are often linked to sudden weight loss (Ministry of Gender and Family, 2024).

The presence of a wasting condition in children poses very high risks to their lives, as immune suppression due to inadequate nutrients leaves them susceptible to illnesses, including infections, which would otherwise be non-life-threatening in healthy individuals (World Bank, 2018). Wasting should, therefore, be diagnosed and managed as soon as possible. From the standpoint of this study, wastings are of great relevance in that their presence could be influenced by the environmental conditions of various districts. Seasonal variation in rainfall and food availability

in various districts of Rwanda vary greatly, thus the need for incorporating these variables into this study (Habyarimana et al., 2016).

2.1.4 Underweight. Being underweight is a form of combined nutritional deficiency, marked by low weight for the given age. A child is considered underweight when his/her z-score of weight-for-age is below -2 of WHO median value (Mukabuterera et al., 2016). Since weight-for-age depends on height and current weight status, being underweight is often defined as a combined indicator reflecting both acute and chronic malnutrition. Therefore, a child may have this type of malnutrition irrespective of whether he/she is stunted, wasted, or both (Akombi et al., 2017).

With regard to Rwanda, the share of underweight children has decreased over recent years due to government intervention aimed at improving the situation in terms of nutrition and expanding access to health services (Mukabuterera et al., 2016). However, at the same time, the current levels of underweight prevalence remain higher than expected based on the general development path of the country, and some inequalities still exist with respect to household wealth and geographic location (Mukabuterera et al., 2016; World Bank, 2018). Earlier studies found that poor households, children of poorly educated mothers, and rural residents were at increased risk for this condition (Habyarimana et al., 2016). In this research, underweight status is taken into consideration alongside stunting and wasting in order to obtain a better understanding of geographic inequalities in malnutrition in 30 Rwandan districts.

2.1.5 Agro-Climatic Indicators. Agro-climatic indicators are quantitative measures employed to characterise the environmental and meteorological conditions that govern agricultural production within a given geographic area; the term itself reflects the integration of climatic science and agricultural systems (World Bank, 2018). These indicators bear directly on child nutritional status because the volume and diversity of food produced at the local level is a primary determinant of household food access (Ministry of Gender and Family, 2024). Under favourable agro-climatic conditions, crop yields tend to be higher, food prices remain relatively stable, and households are more likely to achieve dietary diversity. Conversely, adverse conditions suppress production, elevate food prices, and reduce household dietary quality in ways that can heighten the risk of childhood malnutrition (World Bank, 2018).

2.2 Conceptual Framework

The theoretical foundation for the current research is derived from the UNICEF's framework of child undernutrition (UNICEF et al., 2020) where the factors of malnutrition are organised into three sequentially arranged levels of causality. At the most proximal level, there are immediate causes of malnutrition, such as the lack of proper nutrition and infectious diseases. The immediate causes emerge as consequences of underlying factors, such as lack of food security, poor quality of maternal care and feeding practices, and lack of water and sanitation facilities and health services. Lastly, the more distal basic structural causes include socioeconomic, political and environmental circumstances related to poverty, education, governance, and access to natural resources (Niragire et al., 2025; Habimana and Biracyaza, 2019).

In the context of the current analysis, the agro-climatic variables, as well as socioeconomic factors,

could be interpreted as district-level variables shaping nutritional outcomes among children in Rwanda. Precipitation and NDVI (Normalized Difference Vegetation Index) are considered as measures of environmental productivity affecting food production and availability. On the other hand, socioeconomic factors, such as the household wealth index and education attained by mothers, reflect the ability of families to secure high-quality food products, visit health centers, and ensure proper childcare. Altitude is another variable taken into account since it might influence not only the agricultural potential of a given district, but also the epidemiology of respiratory and infectious diseases, as well as access to services.

The fundamental conceptual assumption that lies at the basis of this study is that there are certain geographical variations in rainfall patterns, environmental productivity, altitude, and socioeconomic status in Rwanda. Consequently, these factors should also lead to different rates of stunting, wasting, and underweight depending on the district. The correlations between these factors are expected to demonstrate spatial autocorrelations due to the similarities in geographic proximity.

2.2.1 Ordinary Least Squares (OLS) Regression. As a methodological baseline prior to the application of spatial models, this study estimates Ordinary Least Squares (OLS) regression equations for each malnutrition outcome and is among the most widely applied statistical techniques for quantifying the linear association between a dependent variable and a set of explanatory covariates, and serves here to establish benchmark estimates of the relationship between district-level malnutrition prevalence and the selected agro-climatic and socioeconomic predictors (Nshimiyiryo et al., 2019).

A foundational assumption of OLS is that observations are mutually independent (Getaneh et al., 1998). This assumption is frequently untenable with geographically referenced data, since outcomes in neighbouring administrative units may be structurally interdependent. Accordingly, following OLS estimation this study subjects the model residuals to a formal test for spatial autocorrelation using Moran's I (Anselin, 1995). Where the residual test yields no evidence of spatial dependence, the OLS specification is regarded as sufficient. Where significant spatial structure persists in the residuals, more appropriate spatially adjusted models are estimated.

2.2.2 Spatial Autocorrelation. Spatial autocorrelation describes the tendency for geographic units in close proximity to exhibit greater similarity in a given attribute than units located far apart (Niragire et al., 2025). Applied to child malnutrition, this concept implies that districts with elevated malnutrition burdens may be systematically adjacent to other high-burden districts, generating clusters of deprivation, while low-burden districts may similarly concentrate in particular sub-regions. Such clustering is termed positive spatial autocorrelation.

Accounting for spatial autocorrelation is analytically important because its presence violates the independence assumption of standard regression models and can produce biased coefficient estimates if ignored. Akombi et al. (2017) study employed the Global Moran's I statistic to formally assess the extent of spatial clustering in malnutrition prevalence across Rwanda's 30 districts. A statistically significant positive value of Moran's I indicates that malnutrition prevalence is distributed in a geographically clustered rather than random pattern.

To pinpoint the specific districts that constitute these clusters, this study additionally computes

Local Indicators of Spatial Association (LISA) (Niragire et al., 2025). LISA assigns each district to one of four cluster categories: High-High hotspots (districts and their neighbours share elevated prevalence), Low-Low coldspots (districts and their neighbours share low prevalence), or spatial outlier categories (High-Low or Low-High), where a district's value diverges markedly from its neighbours. These designations directly support the geographic targeting of nutrition interventions.

2.2.3 Spatial Regression Models. When OLS residuals exhibit significant spatial autocorrelation, spatially explicit regression models become necessary to account for inter-district dependence. Niragire et al. (2025) evaluated two complementary spatial specifications: the Spatial Lag Model and the Spatial Error Model.

The **Spatial Lag Model** incorporates a spatially lagged version of the dependent variable as an additional predictor, thereby modelling a process in which the malnutrition burden of a district is partly determined by the burden in its neighbouring districts. This structure is appropriate where genuine spillover processes operate across administrative boundaries, such as the diffusion of poverty, food insecurity, or health-seeking behaviour.

The **Spatial Error Model** is motivated by a different mechanism: spatial dependence that originates not in the outcome itself but in unmeasured confounders that are geographically organised. Examples include spatially structured cultural food practices, regional infrastructure differences, or common environmental exposures that are not fully captured by the observed predictors.

Selection between the two spatial models is guided by Lagrange Multiplier (LM) diagnostic tests applied to the OLS residuals. The LM framework provides formal guidance on whether spatial dependence is more consistent with a lag or error process, thereby facilitating an evidence-based model selection decision.

2.3 Relevant Studies

2.3.1 Global and Regional Evidence on Child Malnutrition. The global literature on child malnutrition has consistently demonstrated that its consequences extend well beyond the childhood period. According to Victora et al. (2008), poor nutrition in children leads to poor educational attainments, reduced physical strength, and low productivity when they attain adulthood. Furthermore, the study showed that there is an intergenerational aspect in that nutritionally deficient mothers have the highest probability of giving birth to malnourished children and perpetuating the cycle across generations. Such a cyclic mechanism explains why malnutrition remains high in Rwanda despite the country's consistent growth (World Bank, 2018).

At the global policy level, the World Health Organization has put up a goal of reducing childhood stunting by 40% globally before 2025, while the Sustainable Development Goal 2 ensures that signatories commit themselves to eradicating any kind of malnutrition before 2030 (Niragire et al., 2025; Ministry of Gender and Family, 2024). While most other parts of the world have managed considerable progress towards achieving the set goals, sub-Saharan Africa has remained stubbornly resistant to any improvement. Unlike the general global situation, the number of stunted children

increased from about 54.4 million in 2000 to 61.4 million in 2020.

A systematic review by [Akombi et al. \(2017\)](#) synthesising evidence from 49 sub-Saharan African studies identified a consistent constellation of risk factors across the three principal malnutrition indicators. Low maternal educational attainment, limited household economic resources, inadequate access to clean water, low birth weight, male child sex, and rural residence were all independently associated with elevated stunting, wasting, and underweight. Children of adolescent mothers and those from the most economically marginalised households faced the highest multi-indicator burden. Importantly, that review highlighted a systematic gap in the existing literature: most studies addressed only individual- and household-level determinants, overlooking the geographic distribution of malnutrition and failing to examine how environmental conditions such as rainfall variability and vegetation productivity interact spatially with social risk factors.

2.3.2 Studies in Rwanda. Several studies have sought to establish the determinants of childhood malnutrition in Rwanda employing data from the successive waves of the Rwanda Demographic and Health Survey (RDHS). [Habyarimana et al. \(2016\)](#) studied predictors of stunting, wasting and underweight together applying multivariate regression modelling using the RDHS 2010 data. Factors found to contribute to malnutrition included maternal education level, household economic status, safe drinking water access, birth weight, birth order, and the province where the child resided. An interesting result from this study was the high correlation between stunting and being underweight ([Mukabutera et al., 2016](#)), implying the existence of similar determinants and the importance of considering the two indicators jointly.

In the same survey round, [Mukabutera et al. \(2016\)](#) concentrated on determining the factors contributing to being underweight. In particular, boys, multiple births, children of uneducated and underweight mothers, and children from poorer households were at an increased risk of being underweight. Furthermore, the findings revealed that there is generational persistence of underweight as a child's being underweight is associated with a mother's underweight condition. However, while this and other studies provide relevant insights into determinants of malnutrition, they consider the districts merely as a control factor and do not examine the spatial patterns of malnutrition prevalence.

[World Bank \(2018\)](#) extended the analytical scope by drawing on multiple DHS data rounds and documented sustained declines in stunting prevalence in Rwanda, although the burden remained disproportionately concentrated among children from poor and rural households, large families, and mothers with limited education. That analysis additionally noted that relatively few Rwandan children simultaneously benefit from adequate dietary intake, a healthy living environment, and accessible healthcare, three conditions that together underpin healthy child growth. Like its predecessors, however, that analysis neither employed spatial modelling techniques nor incorporated agro-climatic variables.

More recent study has begun to address the spatial dimension of child malnutrition in Rwanda. [Niragire et al. \(2025\)](#) examined spatio-temporal variation in stunting across all 30 districts using data from three RDHS rounds spanning 2010 to 2019/20, applying geo-additive Bayesian models to disentangle district-level spatial effects from individual-level determinants. The analysis revealed significant and temporally evolving spatial effects: the geographic concentration of stunting intensified in the Western and Northern provinces while diminishing in the Eastern

province over the study period. Altitude emerged as a robust and independent predictor across all three survey rounds, with children in districts above 2,000 metres facing consistently higher stunting odds. These findings directly motivate the inclusion of agro-climatic variables in any spatial modelling framework for Rwandan malnutrition.

2.3.3 Environmental and Spatial Factors in Malnutrition. Recognition of the role that environmental and agro-climatic conditions play in shaping child nutritional status has grown substantially in recent years. Rain-dependent agriculture is common in many regions in sub-Saharan Africa; hence, year-to-year variations in rainfall affect child nutritional status because rainfall variability impacts crop yield, food stock availability, and ultimately child nutrition (World Bank, 2018). There is empirical evidence suggesting that districts receiving erratic or low rainfall amounts have high incidences of stunted growth and children being underweight due to low crop yields and consequently low dietary diversities in rural areas in Ethiopia and other sub-Saharan countries (Akombi et al., 2017).

Normalised Difference Vegetation Index (NDVI) is another environmentally related factor that could be included in this analysis due to its impact on child nutrition. The NDVI measure indicates the productivity level of the land area using remote satellite imaging and, therefore, represents agricultural productivity levels and vegetation greenness; that is, higher the NDVI, better is the productivity (World Bank, 2018). Areas with high NDVI have lower rates of food insecurity, and several studies have used NDVI as ecological indicators of food systems' effectiveness in cross-national contexts.

Within Rwanda specifically, Niragire et al. (2025) established altitude as a consistent and statistically significant predictor of child stunting, with risk increasing monotonically with elevation across all three RDHS survey rounds examined. Notwithstanding this evidence, no published study for Rwanda has simultaneously incorporated rainfall, vegetation cover, and altitude within a single spatially explicit analytical framework using DHS-derived malnutrition data. This gap is consequential because Rwanda's five distinct agro-ecological zones generate substantial district-level variation in agricultural capacity that sociodemographic variables alone are unable to capture (Ministry of Gender and Family, 2024).

Collectively, the reviewed literature demonstrates that while the household-, maternal-, and child-level determinants of malnutrition in Rwanda are comparatively well characterised, the geographic and environmental dimensions of the problem remain insufficiently explored. Specifically, no prior study has applied spatial autocorrelation techniques such as Moran's I and LISA to systematically identify malnutrition hotspots across Rwanda's 30 districts, and no study has embedded agro-climatic indicators within a district-level spatial regression framework. The present study addresses both gaps by integrating the 2019/20 RDHS with CHIRPS satellite-derived rainfall data, MODIS NDVI estimates, and altitude measurements, and by applying spatial lag and spatial error models to generate district-level evidence capable of informing Rwanda's national nutrition policy and sub-national intervention targeting (Ministry of Gender and Family, 2024; Niragire et al., 2025).

Rwanda's terrain is marked by considerable topographic and climatic diversity, giving rise to five ecologically distinct agro-zones. The **Congo-Nile Divide** occupying the western ridge is the wettest and coolest zone, supporting a wide range of agricultural activities. The **Central Plateau** records moderate precipitation and carries the highest population density in the country. The **Northern Highlands**, underlain by fertile volcanic soils, rank among the most agriculturally productive areas. By contrast, the **Eastern Savanna** is semi-arid and drought-prone, where rainfall is both lower and more erratic, creating conditions that are inherently less reliable for food production. Finally, the **Bugesera Depression** in the south-east receives the least rainfall of any zone and is especially susceptible to seasonal food shortages.

3.2 Research Design

The study adopts a cross-sectional ecological spatial analysis design, focusing on Rwanda's 30 districts as the unit of observation. This design is well suited to identifying the geographic and spatial structures underlying the distribution of child malnutrition and to assessing how agro-climatic and socio-demographic conditions vary across districts in relation to nutritional outcomes. The district was chosen as the primary unit of analysis because it constitutes the principal administrative level at which Rwanda's health and nutrition policies are planned and implemented.

The analytical workflow unfolds across five sequential phases, as summarised in Figure 3.2: data collection, data cleaning and variable construction, exploratory spatial analysis, spatial regression modelling, and model selection with validation.

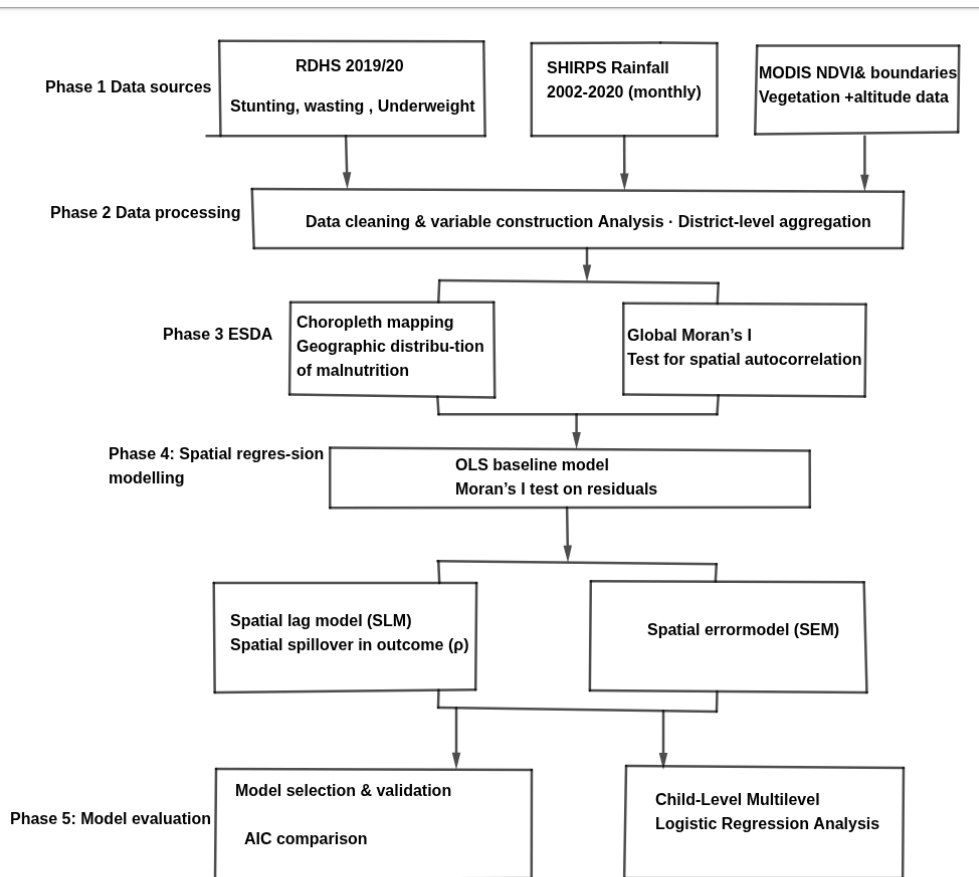


Figure 3.2: Research design flowchart.

3.3 Data Sources

Three complementary data sources were combined to provide comprehensive coverage of child health outcomes, environmental conditions, and the spatial boundaries of Rwanda's 30 districts.

1. **RDHS data:** Child malnutrition indicators were drawn from the 2019/20 Rwanda Demographic and Health Survey (<https://dhsprogram.com>), a nationally representative household survey that records standardised anthropometric z-scores alongside a broad set of child, household, and maternal characteristics. Three data files were utilised: the KR file, containing child-level anthropometric z-scores (HAZ, WAZ, WHZ) and child attributes; the HR file, providing household-level information on wealth, water source, and sanitation access; and the IR file, covering maternal characteristics including educational attainment, body mass index (BMI), and the number of antenatal care (ANC) visits.
2. **Rainfall data:** District-level monthly precipitation (in mm) was sourced from the Climate

Hazards Group Infrared Precipitation with Station data (CHIRPS) product (<https://data.chc.ucsb.edu>), spanning the period 2001–2020. CHIRPS combines geostationary infrared satellite imagery with observations from ground-based rain-gauge networks to generate spatially continuous, high-resolution gridded rainfall fields. Growing-season precipitation statistics were derived from monthly records covering eight months: October, November, and December 2019, and January, March, April, May, and June 2020. February was omitted because it falls within the inter-season dry gap.

3. **Vegetation data:** Land surface vegetation conditions were characterised using the Normalised Difference Vegetation Index (NDVI, ranging from -1 to $+1$) extracted from the Moderate Resolution Imaging Spectroradiometer (MODIS) MOD13A3 product (<https://doi.org/10.5067/MODIS/MOD13A3.061>). NDVI provides a widely used proxy for vegetation vigour and agricultural productivity: values approaching $+1$ indicate dense, actively growing vegetation, while values near -1 denote bare or degraded surfaces. District-level NDVI averages were calculated across the same eight growing-season months, retaining only pixels that passed the MODIS VI_QA quality filter.
4. **Spatial boundary and altitude data:** Administrative boundary shapefiles were obtained from the Rwanda National Institute of Statistics (<https://statistics.gov.rw>) and used to spatially aggregate all variables to a consistent district-level unit. District mean altitude (metres above sea level) was derived from the GPS coordinates of DHS survey clusters and subsequently aggregated to the district level.

3.4 Sample and Variable Construction

The analytical sample comprises children aged under five years drawn from the RDHS dataset. Following data cleaning and file merging, the analysis retains only records with valid anthropometric measurements. Malnutrition status is determined using WHO Child Growth Standards, under which a child is classified as malnourished when their z -score falls below the threshold of -2 standard deviations:

$$Z_i = \frac{AI_i - \mu}{\sigma} \quad (3.4.1)$$

where AI_i denotes the anthropometric measurement for child i , μ is the WHO reference median, and σ is the WHO reference standard deviation (World Health Organization, 2006).

District-level prevalence rates were then computed as survey-weighted proportions to account for the complex, multi-stage sampling design of the RDHS:

$$P_j = \frac{\sum_{i \in j} w_i \cdot \mathbf{1}(Z_i < -2)}{\sum_{i \in j} w_i} \times 100 \quad (3.4.2)$$

where P_j is the weighted malnutrition prevalence (as a percentage) in district j ;

$$w_i = v005_i / 1000000$$

is the normalised DHS sampling weight for child i derived from the **women's individual sample weight** variable; and $1(Z_i < -2)$ is an indicator function taking the value 1 if child i is classified as malnourished and 0 otherwise. The summation extends over all children i residing in district j .

The independent variables considered for the analysis can be categorized into two broad categories: (i) environmental variables that include average rainfall, NDVI, and elevation; and (ii) socio-demographic variables that include the household wealth index, educational attainment of mothers, BMI of mothers, number of ANC visits, quality of water source, and type of toilet facilities. Both household water and sanitation indicators were converted from the HR file as per DHS standards, classifying households based on whether their facilities were improved or unimproved.

3.4.1 Data Cleaning and Handling of Missing Values. Data pre-processing included an examination and treatment of each variable to guarantee validity and consistency before analysis. In the RDHS anthropometric data file, the use of codes 9996, 9997, 9998, and 9999 as sentinel values denotes observations that are not biologically possible and are thus considered missing data. These codes should not be considered as observations since they do not carry any biological meaning. As such, the records coded with these numbers were all treated as missing data. Imputation of z-scores using means was not done to avoid reducing variance, which is necessary for obtaining the true prevalence of malnutrition. Additionally, assigning zero to replace missing data was considered inappropriate since it implied severe malnutrition among the affected children without empirical backing.

For the CHIRPS rainfall dataset, a preliminary audit revealed that 112,800 out of 294,000 observations (approximately 38%) lacked valid precipitation records, leaving 181,200 usable data points (approximately 62% of the total) those null values were mainly caused by pixels lying on areas such as Lake Kivu, where there was no vegetation growth. This was also due to pixels being found at the borders of the country that lay partly within the country's territory. In addition, there were some few monthly data that did not meet the threshold requirement of the CHIRPS during extraction. However, despite the lack of the mentioned data, the available information had enough geographic and temporal resolution to ensure proper district-wise aggregation during the eight-month growing season.

3.5 Agro-Climatic Indicator Construction

Agro-climatic indicators were derived from the CHIRPS and MODIS datasets through the processing steps detailed below.

3.5.1 NDVI. The Normalised Difference Vegetation Index was calculated from the MODIS surface reflectance bands using the standard formula:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{RED}}}{\rho_{\text{NIR}} + \rho_{\text{RED}}} \quad (3.5.1)$$

where ρ_{NIR} and ρ_{RED} denote the surface reflectances in the near-infrared and red spectral bands respectively. The index ranges from -1 to $+1$, with higher values corresponding to denser and healthier vegetation canopies.

The district-level growing-season mean NDVI was obtained by averaging quality-filtered pixel values across the eight selected months:

$$\overline{\text{NDVI}}_j = \frac{1}{K} \sum_{t \in \mathcal{S}} \left[\frac{1}{|\mathcal{K}_j|} \sum_{k \in \mathcal{K}_j} \text{NDVI}_{k,t} \right] \quad (3.5.2)$$

where $K = 8$ is the total number of growing-season months, $\mathcal{S}$ is the set of those months (February excluded due to the dry gap), $\mathcal{K}_j$ denotes the collection of quality-approved MODIS pixels within district j , and $\text{NDVI}_{k,t}$ is the index value at pixel k in month t , retained only when the MODIS VI_QA flag confirms acceptable data quality (Huete et al., 2002).

3.5.2 Rainfall Indicators. Three district-level rainfall indicators were derived from the CHIRPS monthly records over the same eight growing-season months.

The first indicator is the growing-season mean precipitation, calculated as the simple average of monthly totals across the eight months. The second indicator, the **rainfall anomaly**, expresses how the rainfall data season's total compares with the 20-year district baseline:

$$\text{Anomaly}_j = \text{Rain}_j^{(2019/20)} - \frac{1}{20} \sum_{t=2001}^{2020} \text{Rain}_j^{(t)} \quad (3.5.3)$$

where $\text{Rain}_j^{(t)}$ is the growing-season total for district j in year t , computed over the same eight calendar months each year. A positive anomaly signals that 2019/20 was wetter than the historical norm; a negative value indicates a drier-than-usual season. Anomalies were preferred over raw totals because they isolate departures from habitual conditions, which are the deviations most likely to disrupt agricultural output and affect child nutritional status. Although the baseline window (2001–2020) includes the study year, the growing season spans two calendar years; therefore, the complete seasonal total from 2019/20 enters the baseline average, and the anomaly estimates are not materially affected.

The third indicator, the **rainfall coefficient of variation** (CV), quantifies within-season precipitation unpredictability:

$$\text{CV}_j = \frac{\sigma_{\text{monthly},j}}{\mu_{\text{monthly},j}} \times 100 \quad (3.5.4)$$

where $\sigma_{\text{monthly},j}$ and $\mu_{\text{monthly},j}$ are the standard deviation and mean of monthly district rainfall over the eight growing-season months respectively. Higher CV values reflect more erratic intra-

seasonal rainfall distribution, which amplifies agricultural risk and undermines household food security (Cressie, 1993).

3.6 Spatial Weights Matrix

A spatial weights matrix $\mathbf{W}$ is an indispensable component of all spatial statistics and regression models employed in this study, as it formally encodes the neighbourhood relationships among Rwanda's 30 districts. Neighbours were defined using queen contiguity, under which any two districts sharing at least one boundary point or vertex are treated as adjacent (Anselin, 1995). This criterion was preferred over rook contiguity (which demands a shared edge of non-zero length) because queen contiguity also captures corner adjacencies that carry geographic relevance in Rwanda's compact district configuration (LeSage and Pace, 2009). The binary weight between any pair of districts is accordingly defined as:

$$w_{ij} = \begin{cases} 1 & \text{if districts } i \text{ and } j \text{ share any boundary point or vertex} \\ 0 & \text{otherwise} \end{cases} \quad (3.6.1)$$

To ensure comparability across districts with differing numbers of neighbours, the matrix was row-standardised, transforming each spatial lag into a simple weighted average of the values observed in neighbouring units:

$$w_{ij}^* = \frac{w_{ij}}{\sum_{k=1}^n w_{ik}} \quad (3.6.2)$$

Following row-standardisation, each row of $\mathbf{W}$ sums to unity, and the spatial lag $(\mathbf{W}\mathbf{y})_i = \sum_j w_{ij}^* y_j$ yields the neighbourhood-weighted average of the outcome for district i . Applied to Rwanda's 30 districts, the queen contiguity criterion produces an average of approximately 4.6 neighbours per district.

3.7 Exploratory Spatial Data Analysis

Prior to any regression modelling, an exploratory spatial data analysis was undertaken to characterise the geographic distribution of malnutrition indicators and environmental covariates across Rwanda's 30 districts. District-level choropleth maps were produced using the `sf` and `ggplot2` packages in R, with each district shaded according to its observed prevalence or environmental value. This visual step enabled the identification of geographic gradients, inter-district inequalities, and potential spatial clusters before formal statistical testing commenced.

3.8 Global Spatial Autocorrelation (Moran's I)

Following the mapping stage, a formal test was conducted to determine whether the observed geographic patterns in malnutrition prevalence reflect genuine spatial structure or could plausibly arise by chance under spatial randomness. This assessment relied on the Global Moran's I statistic, which provides a single scalar summary of spatial autocorrelation across all 30 districts simultaneously. The statistic is defined as:

$$I = \frac{n}{S_0} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (3.8.1)$$

where:

- $n = 30$ is the total number of districts;
- $S_0 = \sum_i \sum_j w_{ij}$ is the aggregate of all spatial weights, equal to n under row-standardisation;
- w_{ij} is the row-standardised weight between districts i and j as defined in Equation 3.6.2;
- x_i and x_j are the malnutrition prevalence values for districts i and j respectively;
- $\bar{x}$ is the unweighted mean prevalence across all 30 districts.

The numerator accumulates, for every district pair, the product of mean-centred prevalence values weighted by spatial proximity, while the denominator scales this cross-product sum by the overall dispersion of the data, yielding a standardised index of the country-wide spatial pattern (Anselin, 1995). The resulting statistic is interpreted as follows:

- Values close to $+1$ indicate strong positive spatial autocorrelation, whereby high-burden districts cluster near other high-burden districts;
- Values near 0 are consistent with a spatially random distribution of malnutrition prevalence;
- Values approaching -1 indicate spatial dispersion, in which high-burden districts tend to be flanked by low-burden neighbours.

Statistical significance was evaluated via a randomisation-based permutation test; a p-value below 0.05 leads to rejection of the null hypothesis of complete spatial randomness.

3.9 Getis–Ord G_i^* Hotspot Analysis

Global Moran's I shows whether malnutrition is spatially clustered across Rwanda, but it does not show where the clusters are located. To identify districts forming significant hotspots and coldspots, the local Getis–Ord G_i^* statistic was used. This method compares the local sum of values around each district with the overall national pattern and returns a standardised z -score and p -value for each district (Getis and Ord, 1992; Ord and Getis, 1995; Esri, 2024).

$$G_i^* = \frac{\sum_{j=1}^n w_{ij}x_j - \bar{X} \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - (\sum_{j=1}^n w_{ij})^2}{n-1}}} \quad (3.9.1)$$

where x_j is the malnutrition prevalence in district j , w_{ij} is the spatial weight between districts i and j , n is the number of districts, $\bar{X}$ is the mean prevalence across all districts, and S is the standard deviation, defined as:

$$\bar{X} = \frac{\sum_{j=1}^n x_j}{n}, \quad S = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n} - (\bar{X})^2}. \quad (3.9.2)$$

A positive and significant G_i^* z -score indicates a hotspot, meaning that a district is surrounded by districts with high malnutrition prevalence. A negative and significant z -score indicates a coldspot, meaning that a district is surrounded by districts with low malnutrition prevalence. Districts with non-significant values were treated as areas without clear local hotspot or coldspot evidence.

3.10 Spatial Regression Modelling

Classical regression assumes that observations are mutually independent. When data are observed across contiguous geographic units, however, this assumption is routinely violated because neighbouring districts tend to experience similar environmental, socioeconomic, and health conditions. To accommodate this inter-district dependence, three model specifications were evaluated sequentially.

The notation used throughout is as follows. Let $n = 30$ if data aggregate on district level and individual data size if no aggregated performed :

1. $\mathbf{y} = (y_1, y_2, \dots, y_n)'$: the $n \times 1$ vector of district-level malnutrition prevalence rates;
2. $\mathbf{X}$: the $n \times k$ matrix of explanatory variables, encompassing agro-climatic predictors (rain-fall, NDVI, altitude) and socio-demographic covariates (wealth index, maternal education, maternal BMI, ANC visits, water source, and toilet type);
3. $\boldsymbol{\beta}$: the $k \times 1$ vector of regression coefficients;

4. $\mathbf{W}$: the row-standardised spatial weights matrix from Equation 3.6.2;
5. ε : the $n \times 1$ vector of independently and identically distributed errors with mean zero and variance σ^2 .

3.10.1 OLS Baseline Model. An Ordinary Least Squares model was first estimated for each malnutrition outcome to establish a non-spatial benchmark:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (3.10.1)$$

where $\mathbf{y}$ is the (30×1) vector of district prevalences, $\mathbf{X}$ is the design matrix, $\boldsymbol{\beta}$ contains the estimated coefficients, and σ^2 is the error variance. The OLS residuals were subsequently examined for residual spatial autocorrelation using Moran's I to determine whether a spatially adjusted model was warranted.

3.10.2 Spatial Lag Model. The Spatial Lag Model (SLM) is appropriate when the outcome in a given district is partly driven by outcomes in neighbouring districts for instance, when food insecurity or poverty gradients extend across administrative boundaries (Anselin, 1995; Khan and Mohanty, 2018). The model takes the form:

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \varepsilon \quad (3.10.2)$$

where $\mathbf{W}\mathbf{y}$ is the spatially lagged outcome (the neighbourhood-weighted average of malnutrition in adjacent districts) and ρ is the spatial autoregressive coefficient that captures the magnitude of these cross-district spillovers. A statistically significant positive ρ confirms spatial clustering in the outcome; when $\rho = 0$, the specification collapses to standard OLS (Anselin, 1995; Khan and Mohanty, 2018). At the district level:

$$\begin{aligned} y_i = & \rho \sum_{j=1}^n w_{ij}^* y_j \\ & + \beta_1 \text{Rainfall}_i + \beta_2 \text{NDVI}_i + \beta_3 \text{Altitude}_i \\ & + \beta_4 \text{Wealth}_i + \beta_5 \text{Water}_i + \beta_6 \text{Toilet}_i \\ & + \beta_7 \text{Education}_i + \beta_8 \text{BMI}_i + \beta_9 \text{ANC}_i + \varepsilon_i \end{aligned} \quad (3.10.3)$$

3.10.3 Spatial Error Model. The Spatial Error Model (SEM) addresses a different source of spatial dependence: unmeasured confounders that are themselves geographically structured. These include, for instance, culturally embedded food practices that vary by region, spatially concentrated poverty, or shared disease environments (Khan and Mohanty, 2018). In this specification, spatial dependence manifests in the disturbance term rather than in the outcome variable itself:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad (3.10.4)$$

$$\mathbf{u} = \lambda \mathbf{W}\mathbf{u} + \boldsymbol{\varepsilon} \quad (3.10.5)$$

where $\mathbf{u}$ is the $n \times 1$ vector of spatially correlated errors, λ is the spatial error coefficient that quantifies the degree of error dependence across neighbouring districts, and $\boldsymbol{\varepsilon}$ is the residual vector of spatially independent errors remaining after the spatial component has been absorbed (Anselin, 1995). At the district level:

$$\begin{aligned} y_i = & \beta_1 \text{Rainfall}_i + \beta_2 \text{NDVI}_i + \beta_3 \text{Altitude}_i \\ & + \beta_4 \text{Wealth}_i + \beta_5 \text{Water}_i + \beta_6 \text{Toilet}_i \\ & + \beta_7 \text{Education}_i + \beta_8 \text{BMI}_i + \beta_9 \text{ANC}_i + u_i \end{aligned} \quad (3.10.6)$$

where the spatially structured error evolves according to:

$$u_i = \lambda \sum_{j=1}^n w_{ij}^* u_j + \varepsilon_i \quad (3.10.7)$$

3.11 Multilevel Logistic Regression Modelling

District-level spatial models reveal where malnutrition clusters geographically, but they cannot account for the individual-level variability in nutritional status that exists among children residing within the same district. To address this limitation, child-level multilevel logistic regression models were estimated alongside the ecological analyses. The analytical sample comprised 3,785 children nested within 498 DHS survey clusters distributed across Rwanda's 30 districts. District-level environmental variables NDVI, rainfall anomaly, rainfall variability, and altitude were linked to each child according to their district of residence, enabling simultaneous examination of both contextual and individual-level determinants.

The multilevel framework used here follows the hierarchical modelling approaches described by Snijders and Bosker (2012) and Gelman and Hill (2006). Three binary malnutrition outcomes were modelled separately: stunting, wasting, and underweight. For child i residing in cluster j , the model is specified as:

$$\log \left(\frac{p_{ijk}}{1 - p_{ijk}} \right) = \beta_0 + \sum_{q=1}^Q \beta_q X_{qijk} + u_j + v_k \quad (3.11.1)$$

where p_{ijk} is the probability that child i in DHS cluster j and district k is malnourished. The term X_{qijk} represents the predictors included in the model, and β_q is the coefficient for each predictor. The term u_j is the DHS cluster-level random intercept, while v_k is the district-level random intercept. These two terms allow the model to account for unobserved differences between clusters and districts, since children from the same cluster or district may share similar local and geographic conditions.

$$u_j \sim N(0, \sigma_{\text{cluster}}^2), \quad v_k \sim N(0, \sigma_{\text{district}}^2) \quad (3.11.2)$$

Results are reported as odds ratios (OR) with 95% confidence intervals. The proportion of total variance attributable to the cluster level was quantified using the Intraclass Correlation Coefficient (ICC), computed from a null model:

$$\text{ICC}_{\text{cluster}} = \frac{\sigma_{\text{cluster}}^2}{\sigma_{\text{cluster}}^2 + \sigma_{\text{district}}^2 + \pi^2/3} \quad (3.11.3)$$

$$\text{ICC}_{\text{district}} = \frac{\sigma_{\text{district}}^2}{\sigma_{\text{cluster}}^2 + \sigma_{\text{district}}^2 + \pi^2/3} \quad (3.11.4)$$

3.11.1 Model Selection Procedure. Model selection proceeded through a structured sequence. The OLS model (Equation 3.10.1) was estimated first, and its residuals were tested for spatial autocorrelation via Moran's I . Where the test yielded no significant spatial structure, OLS was retained as the preferred specification.

Where significant spatial autocorrelation was detected in the OLS residuals, Lagrange Multiplier (LM) diagnostic tests were applied to distinguish between the SLM and the SEM (Anselin, 1995; Anselin et al., 1996):

$$\text{LM}_{\text{error}} = \frac{\left(\frac{\mathbf{e}^\top \mathbf{W} \mathbf{e}}{\hat{\sigma}^2} \right)^2}{\text{tr}(\mathbf{W}^\top \mathbf{W} + \mathbf{W}^2)} \sim \chi^2(1) \quad (3.11.5)$$

where $\mathbf{e}$ is the OLS residual vector, $\hat{\sigma}^2 = \mathbf{e}^\top \mathbf{e} / n$ is the OLS error variance, and $\text{tr}(\cdot)$ is the matrix trace. When both the standard LM lag (LM_{lag}) and standard LM error (LM_{err}) tests were simultaneously significant, the robust variants (RLM_{lag} and RLM_{err}) were compared, and the model associated with the more significant robust statistic was selected (Anselin, 1995; Anselin et al., 1996).

Final model choice was guided by the Akaike Information Criterion (AIC), which compares candidate models by balancing model fit and model complexity. For a binary logistic model, let $y_i \in \{0, 1\}$ denote the malnutrition status of child i , and let p_i be the predicted probability that the child is malnourished. The likelihood function can be written as

$$L(\theta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}, \quad (3.11.6)$$

where θ represents the model parameters. Taking the natural logarithm gives the log-likelihood:

$\ell(\theta) = \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$ The maximised log-likelihood is obtained after estimating the model parameters, and is written as $\ell(\hat{\theta})$, where $\hat{\theta}$ denotes the parameter values

that maximise the likelihood function (R Core Team, 2026; Taboga, 2021). Therefore, the AIC can be written as

$$\text{AIC} = 2k - 2\ell(\hat{\theta}), \quad (3.11.7)$$

where k is the number of estimated parameters and $\ell(\hat{\theta})$ is the maximised log-likelihood. The model with the lowest AIC was preferred because it provided the best balance between goodness of fit and model simplicity.

4. Results

This chapter presents the findings on child malnutrition in Rwanda using both district-level and child-level analyses. The first part describes the spatial distribution of stunting, wasting, and underweight across the 30 districts. It also examines environmental and socio-demographic predictors, spatial clustering, and district-level regression results and also second part presents child-level multilevel logistic regression models, which were used to test whether district environmental conditions still matter after controlling for child, household, and maternal characteristics.

4.1 Descriptive Statistics

4.1.1 District-Level Malnutrition Summary. the Descriptive statistics were computed to summarise the distribution of malnutrition outcomes across the 30 districts. This helps to interpret the map colour scales and shows the overall level of variation between districts.

Table 4.1: District-level malnutrition summary, Rwanda, RDHS 2019/20

Outcome	Mean (%)	SD (%)	Min (%)	Max (%)
Stunting	33.30	8.38	11.14	50.00
Wasting	1.18	1.30	0.00	5.0
Underweight	7.87	2.76	2.69	14.09

Table 4.1 shows that stunting had the highest district-level burden, with an average prevalence of 33.30% and a maximum value of 50.00%, indicating substantial variation across districts. Wasting was much lower on average, at 1.18%, while underweight had an average prevalence of 7.87%, showing that the three malnutrition outcomes had different levels and spatial variability.

4.1.2 National and District-Level Malnutrition Prevalence. After cleaning the RDHS 2019/20 dataset and removing observations with missing or biologically implausible anthropometric Z-scores, a total of 30 districts with complete data were retained for analysis. Weighted district-level prevalence rates were computed using DHS sampling weights.

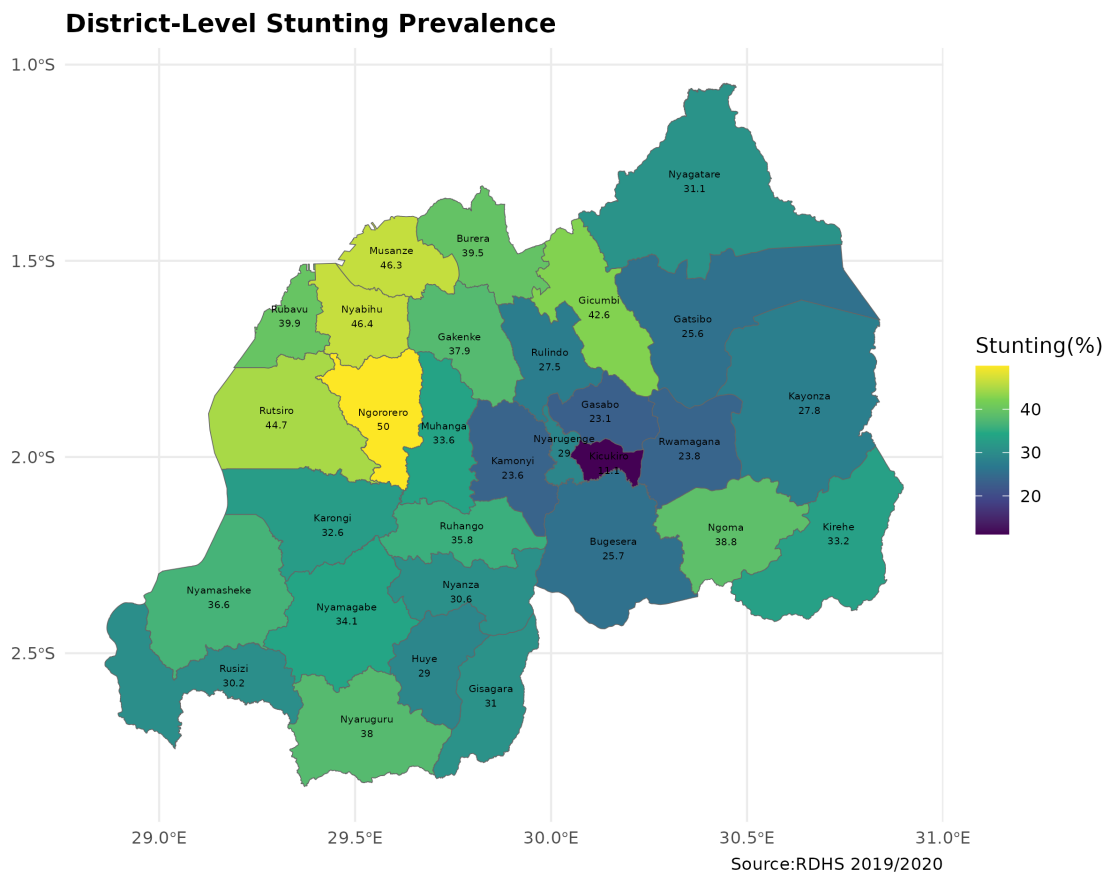


Figure 4.1: District-level stunting prevalence across Rwanda's 30 districts. Source: RDHS 2019/20.

From the Figure: 4.1 that map shows that the highest stunting prevalence is mainly concentrated in the Western and Northern Provinces, especially in districts such as Ngororero have 50% and Lower stunting prevalence is observed around Kigali and some Eastern Province districts, including districts such as Kicukiro with 11.5% this suggests that chronic malnutrition is not evenly distributed, but is more concentrated in highland and north-western districts

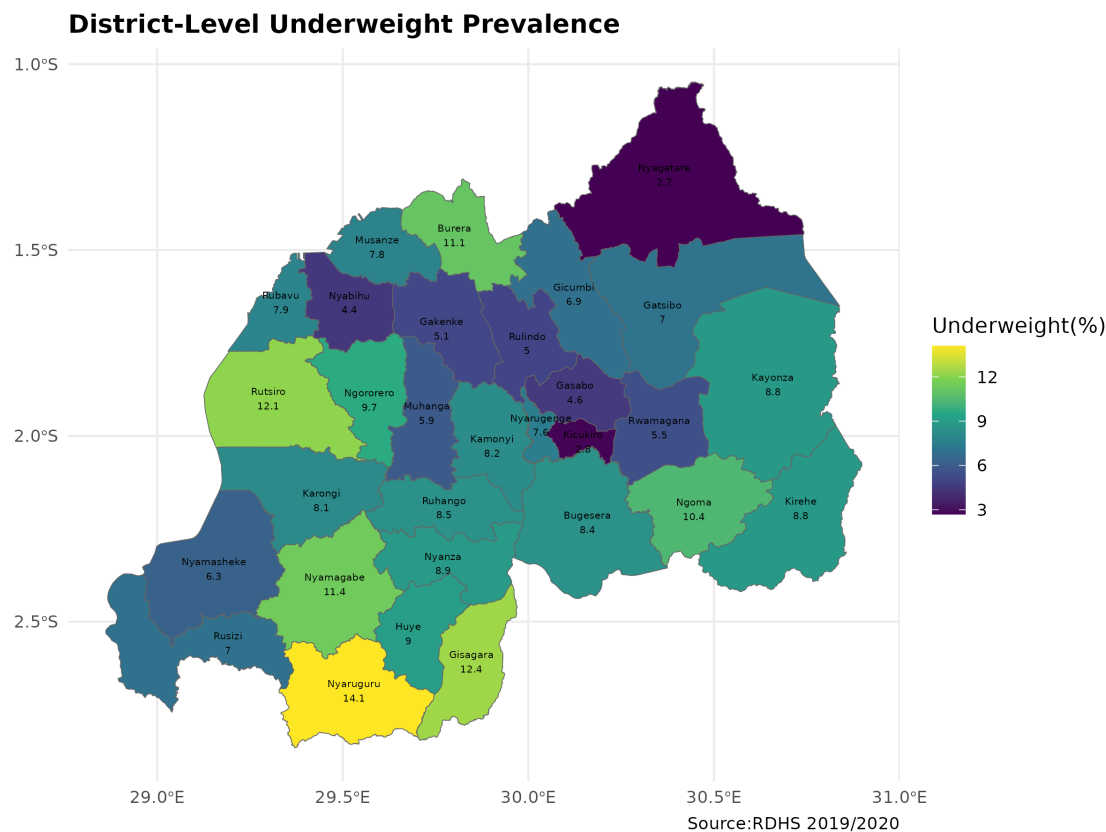


Figure 4.2: District-level underweight prevalence across Rwanda's 30 districts. Source: RDHS 2019/20.

The underweight map in Figure: 4.2 shows an intermediate pattern between stunting and wasting. Higher underweight prevalence is mainly found in the Southern and Western Provinces, especially in districts such as Nyaruguru, Gisagara, Karongi, Rutsiro, Nyamasheke, and some neighbouring districts. Some Eastern Province districts, such as Ngoma and Kirehe, also show relatively high values

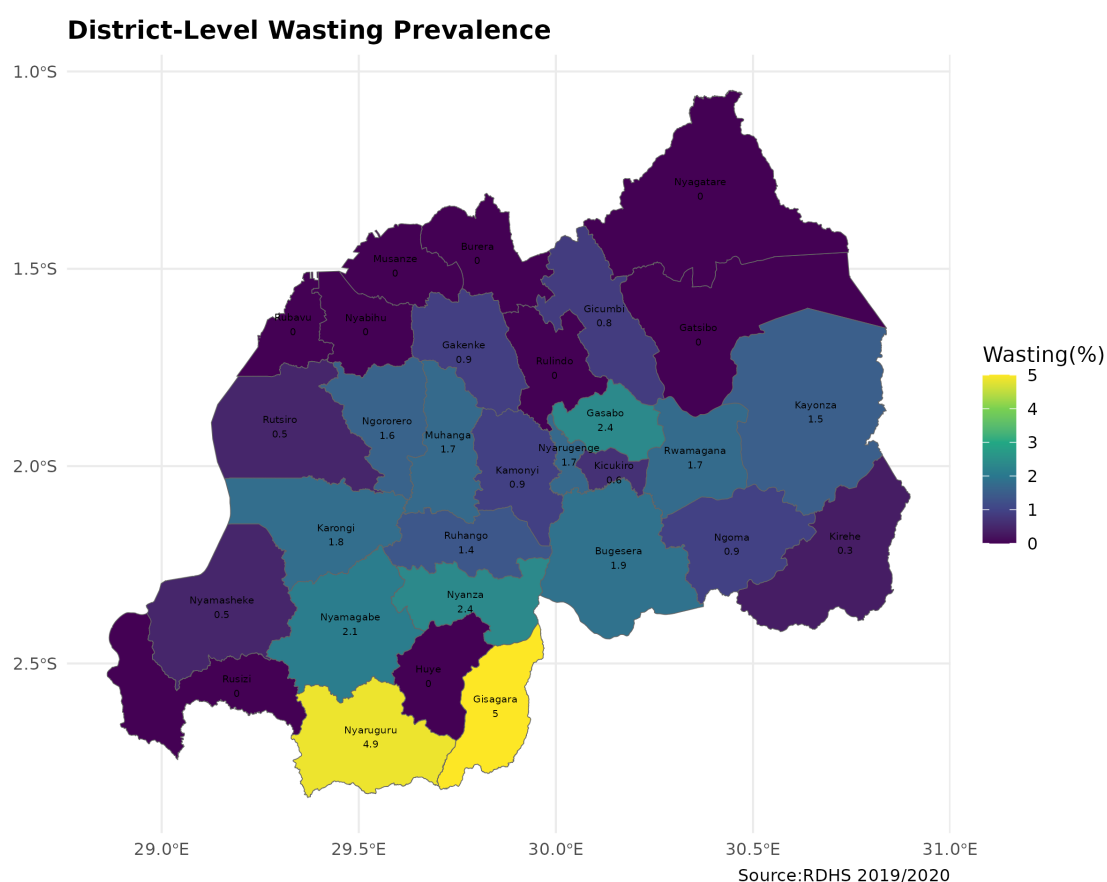


Figure 4.3: District-level wasting prevalence across Rwanda's 30 districts. Source: RDHS 2019/20.

The Figure: 4.3 shows a different pattern from stunting. Wasting prevalence is generally low across most districts, but higher values appear mainly in parts of the Southern Province, especially around Nyaruguru, Gisagara, Huye, and nearby districts. Many districts in the Northern, Western, Eastern, and Kigali provinces show very low or near-zero wasting prevalence. This indicates that wasting is less spatially widespread than stunting

Table 4.2: Environmental predictor summary across 30 districts

Variable	Mean	SD	Min	Max	Unit
Mean NDVI	0.65	0.05	0.52	0.77	Index
Rainfall anomaly	43.68	50.15	-25.73	142.37	mm
Rainfall CV	12.34	4.21	6.50	24.30	%
Mean altitude	1713	243	1160	2650	m asl

4.1.3 Environmental Predictors. The Table 4.2 shows that district-level environmental conditions varied across Rwanda. Mean NDVI ranged from 0.52 to 0.77, showing differences in vegetation conditions, while rainfall anomaly ranged from -25.73 mm to 142.37 mm, indicating that some districts received less rainfall than the baseline and others received more. Altitude also varied substantially, from 1160 m to 2650 m above sea level, which supports the need to consider environmental differences when analysing district-level malnutrition patterns.

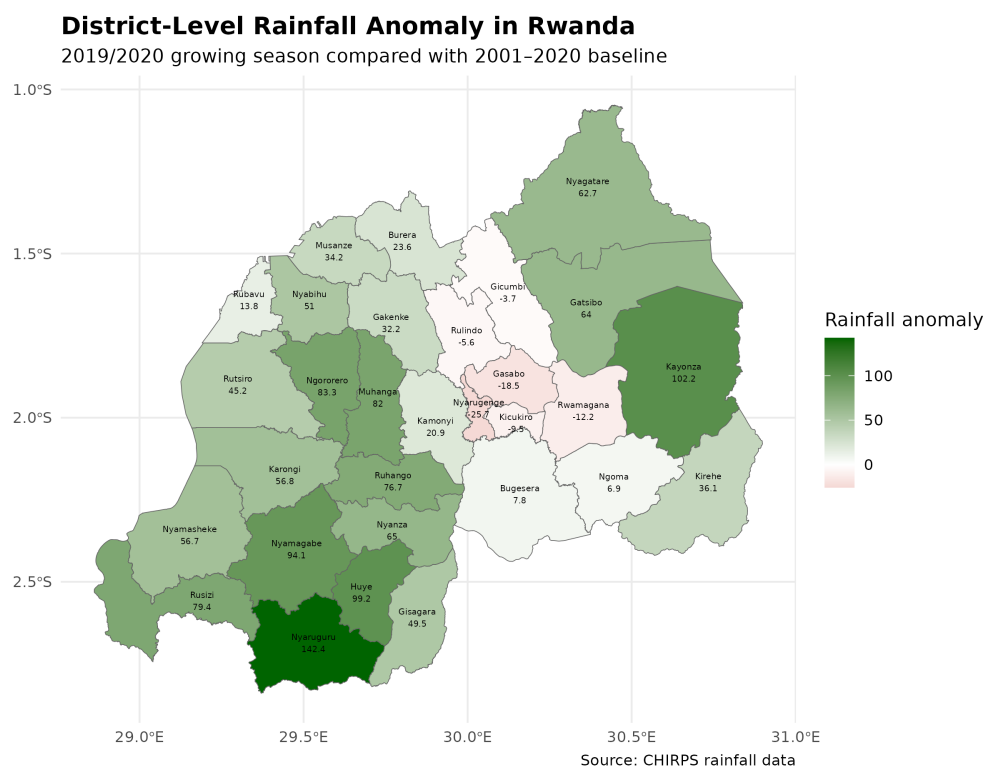


Figure 4.4: District-level rainfall anomaly (mm)

District-level rainfall anomaly from Figure 4.4 it cover 2019/2020 growing season compared with the 2001–2020 baseline, Positive values indicate districts that received more rainfall than the long-term average, while negative values indicate districts that received less rainfall than usual. The map shows that most districts had positive rainfall anomalies, meaning that rainfall was generally above the baseline during the study season.

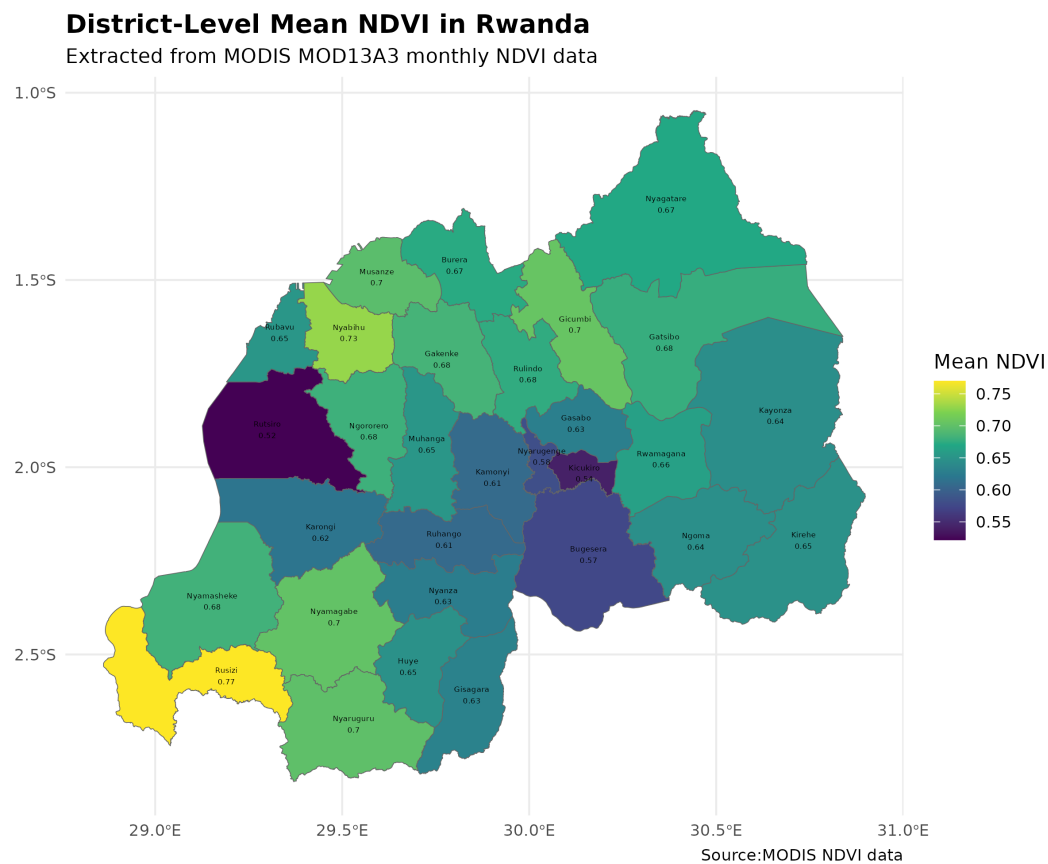


Figure 4.5: District-level mean NDVI during the 2019/20 growing season.

From the Figure 4.5 that map shows the district-level mean NDVI across Rwanda it shows clear spatial variation in vegetation greenness among districts where higher NDVI values are observed mainly in districts such as Rusizi, Nyabihu, Musanze, Nyamagabe, Nyaruguru, and Gicumbi, indicating relatively greener vegetation conditions while the district showed the lower NDVI values are Rutsiro, Kicukiro, Bugesera, Kamonyi, and Ruhango.

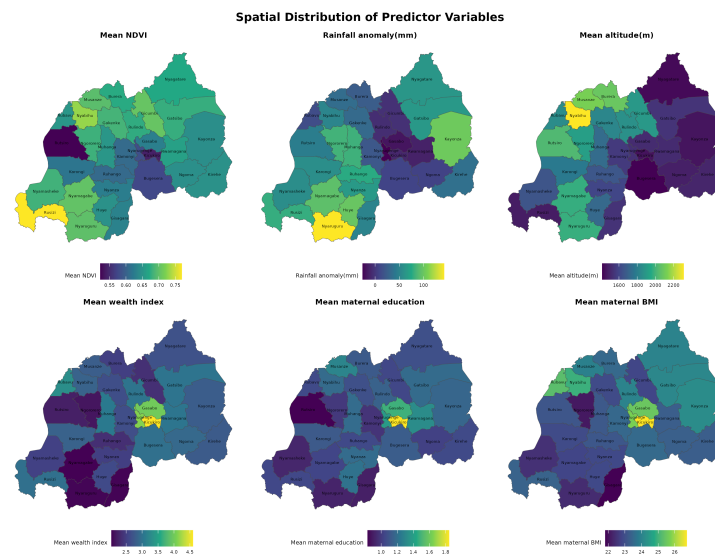


Figure 4.6: Spatial distribution of key predictors across Rwanda's 30 districts: NDVI, rainfall anomaly, altitude, wealth index, maternal education, and maternal BMI.

The rainfall anomaly map shows how rainfall during the 2019/20 growing season differed from the long-term 2001-2020 baseline. Positive values indicate wetter-than-normal conditions, while negative values indicate drier-than-normal conditions. NDVI captures vegetation productivity, which is important because it may reflect agricultural conditions and food availability.

4.2 Exploratory Spatial Data Analysis

Outcome Distributions

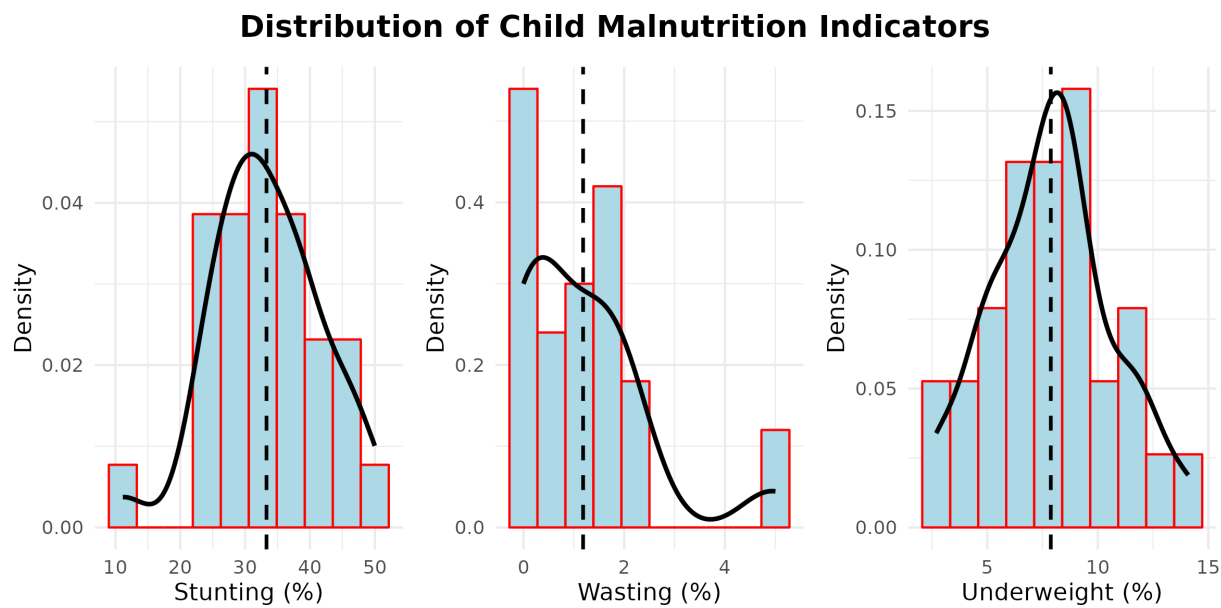


Figure 4.7: Distribution of district-level stunting, wasting, and underweight rates across Rwanda's 30 districts.

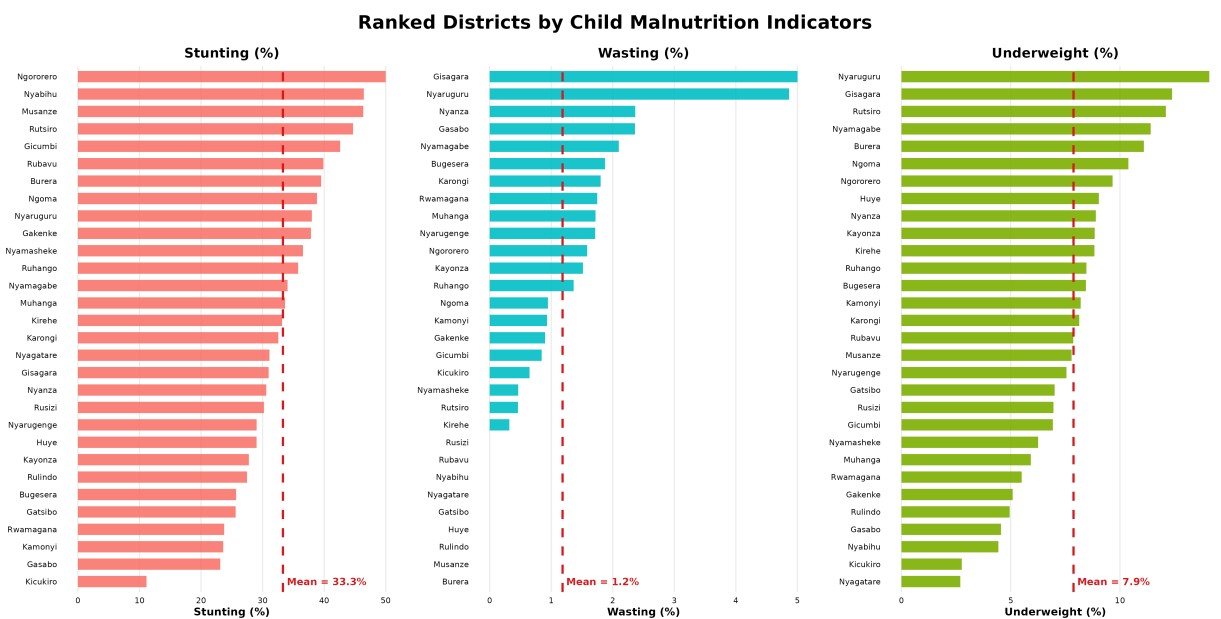


Figure 4.8: Districts ranked by stunting, wasting, and underweight prevalence. The dashed line shows the district mean.

The distribution plots and ranked district bars show that stunting has the highest prevalence across districts, followed by underweight, while wasting remains comparatively low. The ranked

plots also identify districts with unusually high or low prevalence, which helps in understanding priority areas for intervention.

Predictor Correlations

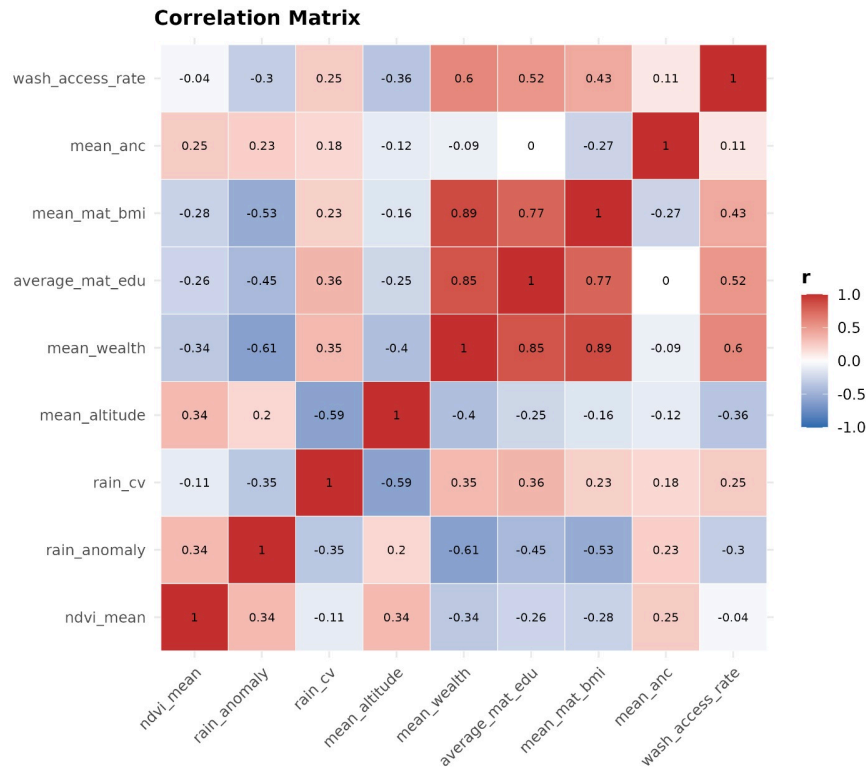


Figure 4.9: Pairwise correlation matrix of candidate predictors

Figure 4.9 shows strong correlation among some predictors. Rainfall in 2019/2020 was highly correlated with baseline rainfall, and household wealth was strongly correlated with maternal education and maternal BMI and Improved toilet access was also strongly related to overall WASH access. These correlations show that some predictors contain overlapping information. For this reason, variable selection is necessary before fitting the final district-level regression models.

Global Spatial Autocorrelation

Table 4.3: Global Moran's I test results for district-level malnutrition prevalence ($n = 30$ districts)

Outcome	Moran's I	p-value
Stunting	0.4447	0.000013
Wasting	0.1782	0.0257
Underweight	0.2331	0.0106

The Moran's I statistics were positive and statistically significant for all three malnutrition indicators. This indicates that malnutrition is spatially clustered across Rwanda rather than randomly

distributed. Stunting showed the strongest spatial autocorrelation ($I = 0.4447$, $p < 0.001$), meaning that districts with high stunting prevalence tend to be located near neighbouring districts that also have high stunting prevalence. Wasting and underweight also showed significant positive spatial clustering, although their spatial dependence was weaker than that of stunting.

Moran Scatter Plots

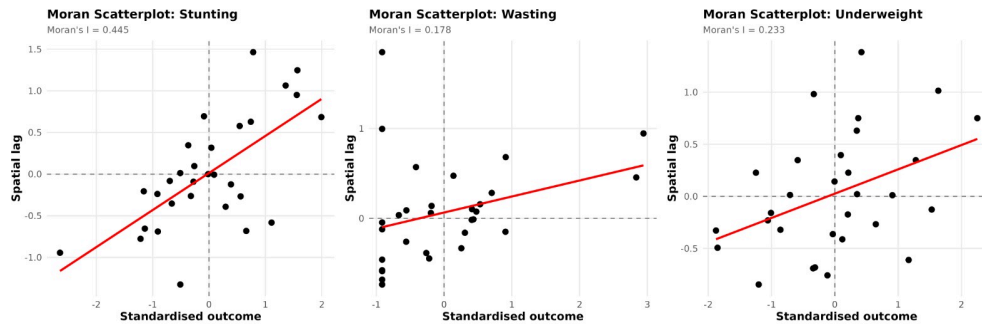


Figure 4.10: Moran scatter plots for stunting, wasting, and underweight

Districts in the upper-right quadrant represent High-High patterns, while districts in the lower-left quadrant represent Low-Low patterns.

4.2.1 G* Hotspot Analysis. Figure 4.11 shows the spatial concentration of malnutrition using the Getis-Ord G_i^* statistic

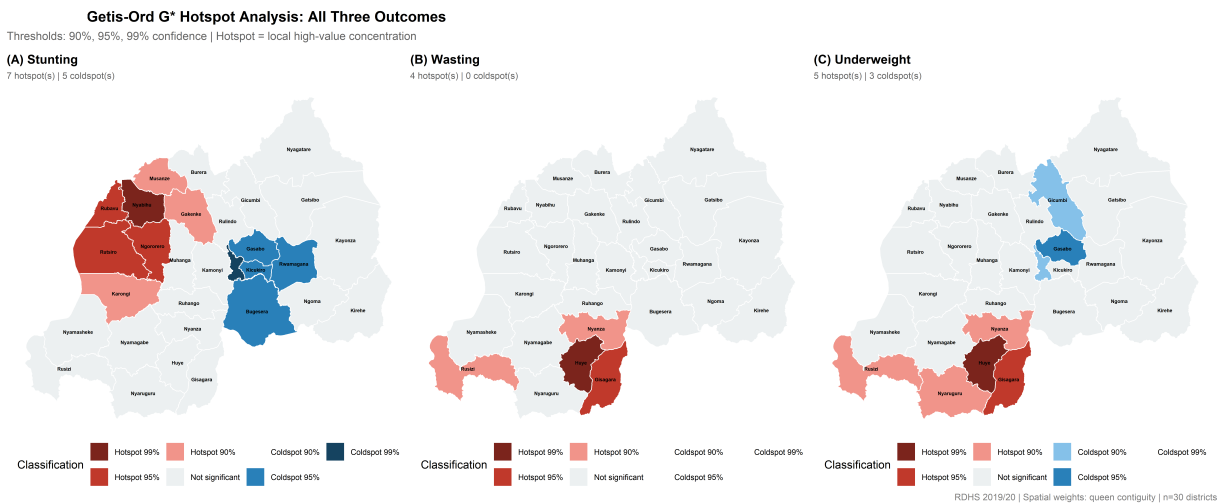


Figure 4.11: Getis-Ord G_i^* hotspot analysis for stunting, wasting, and underweight across Rwanda's 30 districts.

The Moran scatter plots from Figure: 4.10 confirm the positive spatial association found in the Global Moran's I test, means that child malnutrition is geographically clustered rather than ran-

domly distributed across Rwanda. Stunting hotspots were mainly observed in western districts, while coldspots appeared around Kigali and neighbouring districts. Wasting hotspots were concentrated mainly in southern districts, whereas underweight hotspots were also observed in parts of the south. Coldspots for underweight appeared around Kigali and some northern districts. These hotspot patterns help identify districts that may require greater attention for nutrition interventions and resource allocation.

4.3 Variable Selection

Before fitting the district-level regression models, variable selection was conducted using StepAIC and best-subsets regression. This was necessary because the analysis used only 30 districts, so including many predictors could lead to overfitting and unstable estimates. The initial screening results are shown in Table 4.4. For stunting, altitude, maternal education, WASH access, and ANC visits were identified during the screening stage. However, during the final AIC minimisation step, the stunting OLS model was further reduced to two predictors: mean altitude and average maternal education. Therefore, WASH access and ANC visits are not included in the final OLS results in Table 4.5. For wasting, the final model retained NDVI and maternal BMI, while for underweight it retained NDVI, rainfall anomaly, maternal education, WASH access, and wealth index.

Table 4.4: Variables identified during the district-level variable screening stage

Predictor	Stunting	Wasting	Underweight
Mean altitude	Yes	-	-
Mean NDVI	-	Yes	Yes
Rainfall anomaly	-	-	Yes
Average maternal education	Yes	-	Yes
Mean maternal BMI	-	Yes	-
WASH access	Yes	-	Yes
Mean wealth index	-	-	Yes
Mean ANC visits	Yes	-	-

For stunting, altitude, maternal education, ANC visits, and WASH access were selected, suggesting that chronic malnutrition is related to topography, maternal characteristics, and household living conditions. For wasting, NDVI and maternal BMI were selected, indicating that acute malnutrition may be linked to vegetation productivity and maternal nutritional status. For underweight, NDVI, rainfall anomaly, maternal education, WASH access, and wealth were selected, showing that underweight is influenced by both environmental and socio-economic factors.

4.4 Regression Results

OLS Baseline Models

Table 4.5: OLS regression results for stunting, wasting, and underweight

Predictor	Stunting		Wasting		Underweight	
	Coef.	p-value	Coef.	p-value	Coef.	p-value
(Intercept)	17.31	0.066	17.36	0.012	30.16	< 0.001
Mean altitude	0.0217	< 0.001	–	–	–	–
Mean NDVI	–	–	-6.51	0.153	-20.15	0.018
Rainfall anomaly	–	–	–	–	0.0116	0.339
Average maternal edu.	-18.68	< 0.001	–	–	-3.59	0.347
Mean maternal BMI	–	–	-0.505	0.024	–	–
WASH access	0.084	0.12	–	–	0.0327	0.221
Mean wealth index	–	–	–	–	-2.65	0.092
Mean ANC	-5.104	0.15	–	–	–	–
R^2	0.760		0.192		0.550	
Adjusted R^2	0.722		0.132		0.456	
Residual Moran's I	0.0206		0.0730		0.0443	
Moran p-value	0.315		0.167		0.243	

Table 4.5 presents the final OLS models after AIC-based reduction. The stunting model had the strongest explanatory power, with $R^2 = 0.760$ and adjusted $R^2 = 0.721$, showing that mean altitude and average maternal education explained a large share of the between-district variation in stunting. Both predictors were statistically significant at $p < 0.001$.

In contrast, the wasting model had a low $R^2 = 0.192$, meaning that only 19.2% of the variation in wasting prevalence was explained by the selected predictors. This suggests that wasting was less well explained by the district-level environmental and socio-demographic variables used in this study, possibly because wasting is more acute and may depend on short-term household or local community conditions.

The residual Moran's I tests were not statistically significant for stunting ($p = 0.315$), wasting ($p = 0.167$), and underweight ($p = 0.243$), this means that, after fitting the OLS models, there was no strong evidence of spatial autocorrelation remaining in the residuals. In other words, the selected predictors explained most of the spatial structure that was observed before modelling. Therefore, the OLS models were considered adequate, and there was limited statistical evidence that spatial lag or spatial error correction was needed.

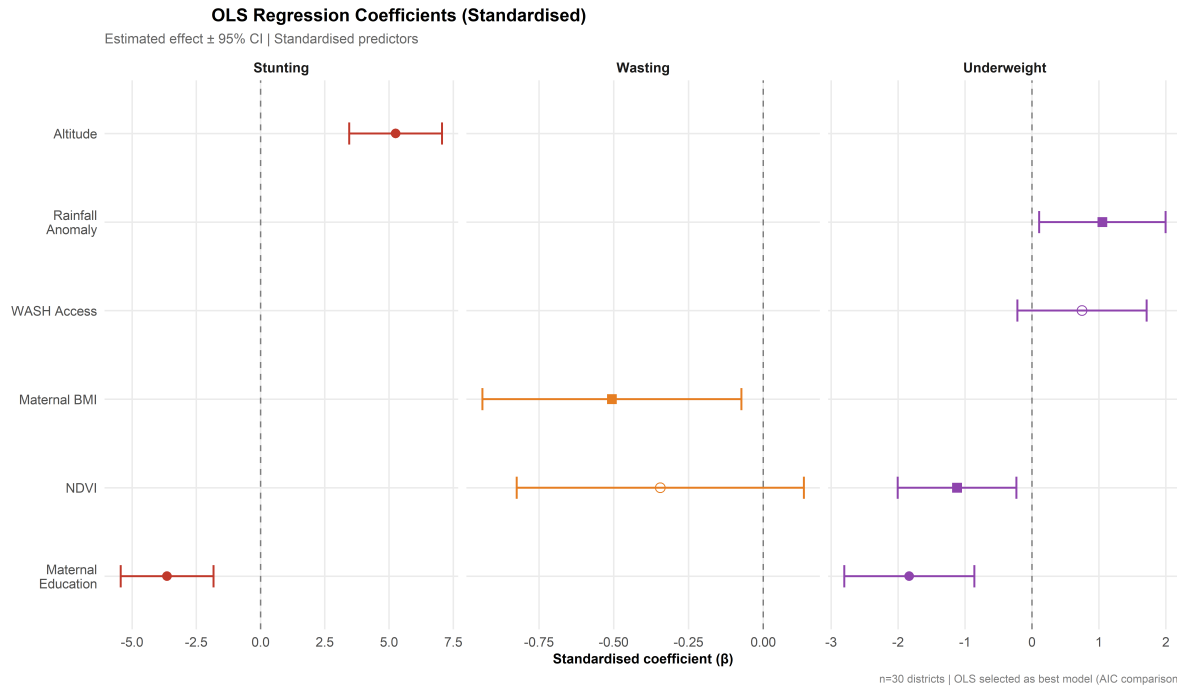


Figure 4.12: Standardized OLS regression coefficients with 95% confidence intervals for district-level malnutrition outcomes.

Figure 4.12 summarizes the effects of selected predictors on district-level malnutrition outcomes. Higher altitude was associated with increased stunting, while maternal education reduced stunting prevalence. Maternal BMI showed a protective effect against wasting, and NDVI was associated with lower underweight prevalence.

Spatial Lag and Spatial Error Models

Table 4.6: Spatial parameter estimates and AIC values for spatial lag and spatial error models

Outcome	Model	Parameter	p-value	AIC
Stunting	Spatial lag	$\rho = 0.142$	0.432	182.76
	Spatial error	$\lambda = 0.049$	0.861	183.35
Wasting	Spatial lag	$\rho = 0.208$	0.383	102.57
	Spatial error	$\lambda = 0.171$	0.537	102.95
Underweight	Spatial lag	$\rho = 0.125$	0.578	136.82
	Spatial error	$\lambda = 0.130$	0.674	136.95

Table 4.6 contain all three outcomes, the spatial lag parameter ρ and the spatial error parameter λ were not statistically significant. This means, after including the selected environmental and socio-demographic predictors, there was no strong evidence of remaining spatial dependence requiring spatial correction.

4.5 Model Selection and Validation

Table 4.7: AIC comparison of OLS, spatial lag, and spatial error models

Outcome	Model	AIC	Δ AIC
Stunting	OLS	181.38	0.00
	Spatial lag	182.76	1.38
	Spatial error	183.35	1.97
Wasting	OLS	101.33	0.00
	Spatial lag	102.57	1.24
	Spatial error	102.95	1.62
Underweight	OLS	135.13	0.00
	Spatial lag	136.82	1.69
	Spatial error	136.95	1.82

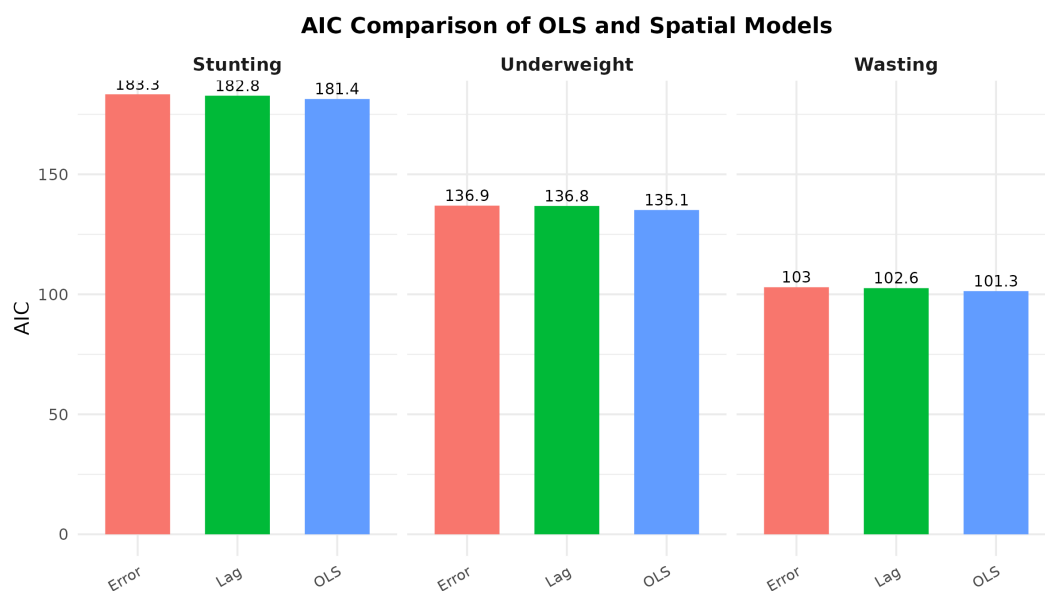


Figure 4.13: AIC comparison of OLS, spatial lag, and spatial error models for stunting, wasting, and underweight.

The AIC comparison shows that the OLS model had the lowest AIC for all three outcomes. Although the spatial lag and spatial error models were also estimated, their AIC values were higher and their spatial parameters were not statistically significant. Therefore, the OLS models were selected as the final models.

Table 4.8: Final selected model for each malnutrition outcome

Outcome	Final model	Key predictors	Adj. R^2	Residual Moran's I
Stunting	OLS	Altitude, maternal education	0.722	0.0206 ($p = 0.315$)
Wasting	OLS	NDVI, maternal BMI	0.132	0.0730 ($p = 0.167$)
Underweight	OLS	NDVI, rainfall anomaly, maternal education, WASH access, wealth	0.456	0.0443 ($p = 0.243$)

Overall, the results show that different factors explain different dimensions of child malnutrition at the district level in Rwanda. Stunting, as a chronic condition, is mainly associated with altitude and maternal education. Wasting, as an acute condition, is more closely related to maternal nutritional status, with NDVI showing a negative but non-significant association. Underweight, as a composite indicator, is strongly related to vegetation productivity and socio-economic conditions. The absence of significant spatial autocorrelation in the OLS residuals suggests that the selected predictors adequately explain the main geographic structure of malnutrition across Rwanda's 30 districts.

4.6 Child-Level Multilevel Logistic Regression Analysis

Child-level multilevel logistic regression models were fitted using 3,785 children from the RDHS 2019/20 dataset. Each child was linked to district-level environmental conditions, including NDVI, rainfall anomaly, rainfall variability, and altitude. This made it possible to examine whether district environmental conditions were associated with child malnutrition after controlling for child, household, and maternal characteristics.

The models included random intercepts for both district and DHS survey cluster. This was important because children from the same district or survey cluster may share similar environmental, household, and community conditions.

Four nested models were estimated. M0 was the null model with only district and cluster random intercepts. M1 added child-level factors, including child age, sex, birth order, and birth interval. M2 added household and maternal factors, including household wealth, maternal education, maternal BMI, antenatal care visits, and WASH access. M3 was the full model and added district-level environmental variables: NDVI, rainfall anomaly, rainfall variability, and altitude. The comparison between M3 and M2 was used to test whether environmental factors added explanatory power beyond child, household, and maternal characteristics.

Variance Decomposition and ICC Results

Table 4.9: Intraclass correlation coefficient results from the null multilevel model

Outcome	District ICC (%)	Cluster ICC (%)	Child-level share (%)
Stunting	2.76	6.35	90.89
Wasting	0.00	84.17	15.83
Underweight	0.00	8.09	91.91

Table 4.9 shows that most variation in stunting and underweight occurred at the individual child level. The district ICC for stunting (2.76%) fell below the conventional 5% threshold, indicating minimal variance between-districts in individual stunting status. However, district random intercepts were retained to reflect the 3-level DHS sampling structure and to allow comparison with cluster-level effects. Sensitivity analyses with only cluster-level random effects produced similar fixed-effect estimates (not shown)

For wasting, the district ICC was 0%, while the cluster ICC was very high (84.17%). This means that wasting was not mainly explained by broad district-level differences, but by local survey-cluster conditions. Therefore, wasting should not be interpreted as a strong district-environmental outcome in this analysis.

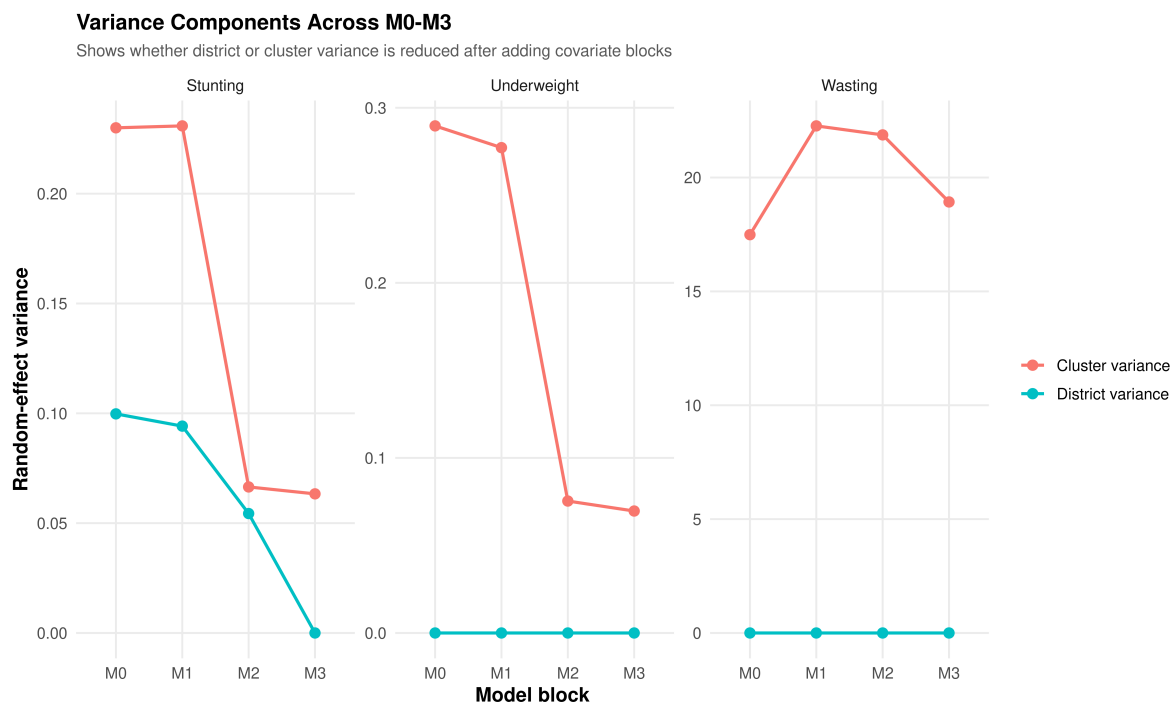


Figure 4.14: Variance components across M0–M3 multilevel models

Figure 4.14 shows how the random-effect variance changed after adding each block of predictors. For stunting, district-level variance declined after adding environmental variables in M3, suggesting that environmental conditions explained part of the geographic difference in stunting.

For wasting and underweight, district-level variance was close to zero across the model blocks. This confirms that there was little district-level variation left to explain for these two outcomes.

4.6.1 Contribution of Environmental Factors. Table 4.10 shows that environmental variables significantly improved the stunting model ($p < 0.001$)

Table 4.10: LRT: M3 (with district environmental predictors) vs M2 (household and maternal predictors only)

Outcome	Environmental LRT p-value	District ICC (%)	Cluster ICC (%)
Stunting	0.000079	2.76	6.35
Wasting	0.752800	0.00	84.17
Underweight	0.733000	0.00	8.09

This means that NDVI, rainfall anomaly, rainfall variability, and altitude added useful information for explaining stunting after controlling for child, household, and maternal characteristics. For wasting and underweight, the environmental block was not statistically significant. This means that the district-level environmental variables did not add clear explanatory power for these two outcomes once child, household, and maternal factors were controlled.

Environmental Effects from the Final M3 Model

Table 4.11: Environmental predictors from the final M3 multilevel model

Outcome	Predictor	OR	95% CI	p-value
Stunting	NDVI	1.075	0.986–1.172	0.100
Stunting	Rainfall anomaly	0.908	0.831–0.992	0.032
Stunting	Rainfall CV	0.926	0.836–1.025	0.139
Stunting	Altitude	1.176	1.063–1.301	0.002
Underweight	NDVI	0.907	0.787–1.045	0.177
Underweight	Rainfall anomaly	1.048	0.906–1.212	0.526
Underweight	Rainfall CV	1.007	0.846–1.198	0.942
Underweight	Altitude	1.024	0.861–1.217	0.792

In the stunting model, rainfall anomaly was negatively associated with stunting ($OR = 0.908$, $p = 0.032$), meaning that children in districts with higher rainfall anomaly had lower odds of stunting. Altitude was positively associated with stunting ($OR = 1.176$, $p = 0.002$), showing higher stunting risk in higher-altitude districts.

For underweight, none of the environmental predictors were statistically significant. This suggests that underweight was more strongly related to household, maternal, and child-level factors than to district environmental conditions.

Note: Wasting is not included in Table 4.11 because the environmental block was not significant and the district ICC was 0%. Therefore, the wasting environmental coefficients should not be

used as final evidence of district-level environmental effects

4.6.2 Final Multilevel Odds Ratio Results. Figure 4.15 shows that stunting was affected by both environmental and socioeconomic factors.

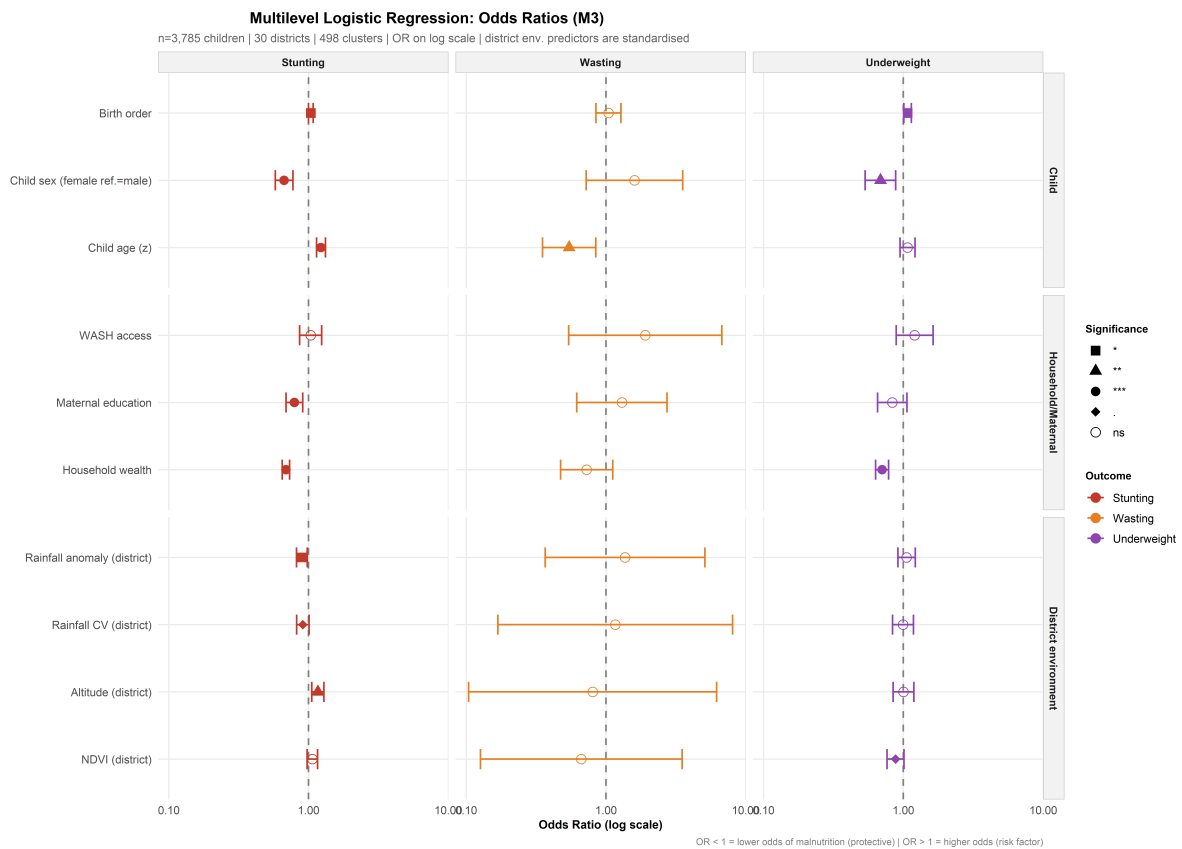


Figure 4.15: Odds ratios from the final child-level multilevel logistic regression models

Higher altitude increased the odds of stunting, while rainfall anomaly, household wealth, maternal education, ANC visits, and female child sex were protective. Child age increased the odds of stunting.

For wasting, the main evidence points to local cluster-level conditions rather than district environmental factors. Maternal BMI and child age were the clearest protective factors. For underweight, household wealth, maternal BMI, ANC visits, and female child sex reduced the odds, while higher birth order increased the odds.

4.6.3 Model Validation. Model validation was conducted using a cluster-level split which was reproduced using fixed random seed to ensure that the results are reproducible, where 80% of DHS clusters were used for training and the remaining 20% were used for testing. This produced 3,026 children in the training set and 759 children in the test set. The split was done at cluster level instead of child level because children from the same DHS cluster may share similar community, household, and environmental conditions; therefore, putting children from the same cluster in both training and testing data could make the model look better than it really is.

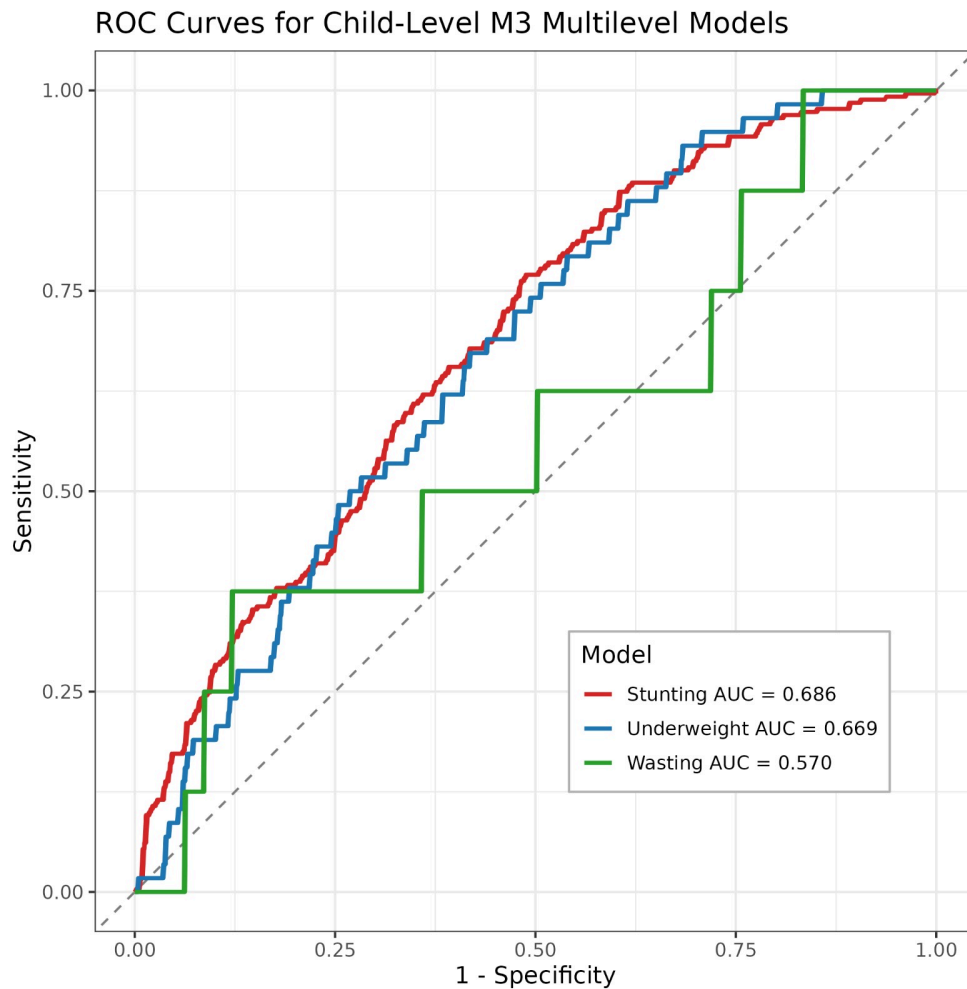


Figure 4.16: ROC curves for child-level M3 multilevel models

Figure 4.16 presents the ROC curves for the final child-level multilevel models. The diagonal dashed line represents random classification, while curves above this line show that the model has some ability to distinguish children with malnutrition from those without malnutrition. The stunting model had the highest AUC of 0.686, followed by the underweight model with an AUC of 0.669. These values indicate moderate discrimination, meaning that the selected child, household, maternal, and environmental predictors helped the models separate children with and without stunting or underweight, although the predictive strength was not very high.

The wasting model had the lowest AUC of 0.570, which shows weak discrimination. This result should be interpreted carefully because wasting is less common in the data compared with stunting and underweight. When an outcome is rare, the test set contains fewer positive cases, so the ROC curve can become less stable and more step-like, as shown in the wasting curve. In this situation, ordinary accuracy may be misleading because a model can classify most children as not wasted and still appear accurate, even when it fails to identify wasted children well.

Overall, the validation results show that the final multilevel models performed better for stunting

and underweight than for wasting. This is reasonable because stunting and underweight are more likely to be linked with longer-term child, household, maternal, and environmental conditions, while wasting is more acute and may depend on short-term illness, recent food shortage, infection, or local shocks that were not fully captured by the available variables. Therefore, the wasting model should be discussed more cautiously, and its performance should be judged using measures such as sensitivity, balanced accuracy, or precision-recall results in addition to ROC-AUC.

5. Conclusion and Recommendations

5.1 Conclusion

The results from Table: 4.3 showed that child malnutrition was spatially clustered across Rwanda, but the strength of clustering differed by outcome. Stunting had the strongest spatial pattern, with Global Moran's $I = 0.4445$ and $p < 0.001$. Wasting also showed positive spatial clustering, but it was weaker ($I = 0.1782$, $p = 0.0257$), while underweight showed moderate clustering ($I = 0.2331$, $p = 0.0106$). This means that districts with high malnutrition were not randomly distributed, especially for stunting. The maps also showed that stunting was more concentrated in western and northern highland districts, wasting was generally low and more scattered, and underweight was higher mainly in southern and western districts.

The district-level OLS results showed that the selected predictors explained stunting and underweight better than wasting. The stunting model had the strongest explanatory power, explaining 76.0% of the between-district variation in stunting prevalence ($R^2 = 0.760$), while the underweight model explained 55.0% ($R^2 = 0.550$). In contrast, the wasting model explained only 19.2% of district variation ($R^2 = 0.192$), suggesting that wasting was not well captured by district-level environmental and socio-demographic factors. After fitting the OLS models, residual Moran's I tests were not statistically significant for stunting ($p = 0.315$), wasting ($p = 0.167$), and underweight ($p = 0.243$), indicating that no strong spatial autocorrelation remained in the residuals and that the OLS models were sufficient compared with spatial lag or spatial error models. However, the strong district-level fit for stunting should not be interpreted as strong individual-level prediction, because the OLS model explains variation between districts, while the child-level validation evaluates prediction among individual children. This explains why altitude and maternal education explained a large share of district-level stunting variation, but the child-level multilevel model achieved only moderate discrimination with an AUC of 0.686. This difference reflects the risk of ecological fallacy, where relationships observed at district level may not fully translate into accurate prediction at child level.

The results child-level multilevel showed that district environmental factors were most important for stunting. Children living in districts with better rainfall conditions had lower odds of stunting, while children living in higher-altitude districts had higher odds of stunting. Specifically, rainfall anomaly reduced the odds of stunting ($OR = 0.908$, $p = 0.032$), while altitude increased the odds of stunting ($OR = 1.176$, $p = 0.002$). However, NDVI did not show a clear protective effect for stunting at the child level. At the district level, better vegetation conditions were more clearly linked to lower underweight prevalence, but this effect was not strong in the child-level model after controlling for household, maternal, and child factors .

According to Figure 4.16, the most discriminative power is associated with the stunting model with an AUC of 0.686, followed by the underweight model with an AUC of 0.669 . These results suggest that the covariates of the child, household, mother, and district level are reasonably good at distinguishing between children suffering from stunting or underweight and children not suffering from stunting or underweight. The model of wasting, on the other hand, has very poor

discriminative power because of the low value of its AUC (0.570). However, this result should be taken into consideration carefully, because wasting is less common than stunting or underweight, making it difficult to validate the latter model since only a small proportion of children have been diagnosed with wasting. There is less signal than noise with a limited amount of wasted children, leading to a low number of true positives, whereas a high number of false negatives would be present with accuracy measures in such an imbalanced case. Stunting and underweight may be more easily distinguished than wasting using the selected predictors, probably owing to short-term diseases, food shortage, or even local events affecting only a few people at a particular moment.

5.2 Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. **Geographically targeted nutrition interventions.** Nutrition programmes should prioritise districts identified as hotspot areas, especially areas with high stunting prevalence. Prevalence maps should also be compared with population burden maps so that resources are directed to both high-rate and high-population districts.
2. **Altitude-sensitive food security programming.** Nutrition-sensitive agricultural programmes should be strengthened in high-altitude and agro-climatically vulnerable districts, where environmental conditions may contribute to chronic child undernutrition.
3. **Climate-sensitive nutrition planning.** District nutrition planning should include climate and environmental indicators such as rainfall anomalies, rainfall variability, vegetation productivity, and altitude, especially for monitoring stunting risk.
4. **Maternal and household support.** Maternal education, antenatal care, household welfare, and WASH-related interventions should be strengthened, since these factors were linked with child malnutrition in the analysis.
5. **Community-level surveillance for wasting.** Wasting should be monitored closely at community level because it was weakly predicted by district-level environmental variables and may depend more on short-term illness, food shortage, or local shocks.

5.3 Limitations

Several limitations should be acknowledged. The RDHS 2019/20 dataset is cross-sectional, so this study can identify associations between environmental conditions and child malnutrition, but it cannot prove direct causal relationships.

The district-level analysis was also limited by the small number of districts in Rwanda, since only 30 districts were available for ecological modelling. This reduced statistical power and may have made some coefficient estimates less stable.

Another limitation is that rainfall, NDVI, rainfall variability, and altitude were aggregated at district level. This means that all children within the same district were assigned the same environmental values, even though environmental conditions may differ within districts.

Wasting was also rare in the cleaned sample, which made its model estimates and validation results less stable. This may explain why the wasting model had weaker predictive performance compared with stunting and underweight.

Finally, the study depended on available secondary data, so some important factors such as detailed child feeding practices, recent illness, soil moisture, land surface temperature, and crop productivity were not fully captured.

5.4 Future Work

Future studies could improve this work by using longitudinal data to better examine how changes in climate and household conditions affect child malnutrition over time.

Future work could also include more detailed environmental indicators, such as land surface temperature, soil moisture, drought indices, and crop productivity measures, to better explain the pathways between agro-climatic conditions and child nutrition.

Another important direction is to analyse stunting, wasting, and underweight together using the **Composite Index of Anthropometric Failure (CIAF)**. This method combines the three malnutrition outcomes into one broader measure of child nutritional failure, while still showing which form of malnutrition is most dominant. This would help future studies identify whether stunting, wasting, underweight, or their combinations contribute most to the overall burden of child malnutrition.

Future studies could also use methods such as principal component analysis or latent class analysis to group children with similar malnutrition patterns and identify the dominant type of nutritional risk in different districts.

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